



UnionPay Integrated Circuit Card Specifications

— Basic Specifications

Part I Debit/Credit Application Overview



Version 2017

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Summary of Revisions

The changes listed below are associated with the current version.

Change Description	Revised/ Deleted/ Added	Section
Revised the description of ADF file to clarify the indication of the DF name of ADF file.	Revised	3.1.1
Switched the order of Section 4.1.11 and 4.1.12, and updated the transaction process order in Figure 1.	Revised	4.1.11, 4.1.12
Revised Figure 2 Flow chart of Application Selection Processing, when the application is blocked, the card returns FCI and 6283 in SELECT response.	Revised	4.2.1.7
Revised the description about “CVM not required”.	Revised	4.2.6.5.3.4
Updated the transaction process order in Figure 16: switched the order of “Terminal processes Issuer Script” and “Terminal completes transaction”.	Revised	4.2.11
Deleted the note of Table 41 about reversal transactions sent by the Terminal for it is outside the scope of UICS.	Deleted	4.2.11.5
Updated the reference from “Part III Appendix A” to “Part II Appendix A”, and “Part III” to “EMV 4.3 Book 3”	Revised	All

1 Application Scope

This book applies to all UPI Participants.

2 References

The clauses of the following documents accessed by *UnionPay Integrated Circuit Card Specifications - Basic Specifications* are considered clauses of this book. The modification list (excluding corrected contents) or revised edition attached to the dated documents shall not apply to this book. However, Participants may study whether to apply the latest versions of such documents. The latest versions of undated references shall apply to this book.

EMV Integrated Circuit Card Specifications for Payment Systems, Version 4.3 Book 1
Application Independent ICC to Terminal Interface Requirements

EMV Integrated Circuit Card Specifications for Payment Systems, Version 4.3 Book 3
Application Specification

ISO/IEC 7816-4 Identification cards--Integrated circuit(s) cards with contacts Part IV
Interindustry exchange commands

ISO/IEC 7816-5 Integrated circuit(s) cards with contacts -- Part 5: Numbering system
and registration procedure for application identifiers

3 File, Data Element, and Data Object List

3.1 File Structure

The organization structures of the files in this Specification are from and comply with the basic organization structure in ISO 7816-4.

This section describes the application file structure that complies with this specification.

From the terminal's perspective, the files on the IC card are in a tree structure. Every branch of the tree is an application definition file (ADF). An ADF is an entry of one or multiple application elementary files (AEF). An ADF and the relevant data files are on the same branch of the tree.

The data objects of the data files in the IC card that can be read /written are stored as records. The structures of file and the means of reference, described below, depend on the usages of the files. With the exception of the directory file that will be described in the next section, the layout of other data files that can be read/ written via an IC card should be defined by the Issuer.

3.1.1 ADF

The tree structure of ADF:

- Connects data file with applications.
- Ensures the independence of applications.
- Achieves access to logical structure via application selection.

From the view of the terminal, an ADF is just a file that contains data objects encapsulated in its file control information (FCI). Please refer to Table B.27 of *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification* for details on data objects encapsulated by FCIs.

The Dedicated File (DF) name of ADF file represents the applications that an IC card supports. It can be an AID, or begin with an AID, and appended with a suffix to make up a file name.

3.1.2 AEF

AEF, with short file identifier (SFI) range of 1-10, includes a basic data object or multiple "basic encoding rule - tag length value" (BER-TLV) data objects. For data objects encoded in BER-TLV see Appendix B, *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification*. Once it is selected, the AEF with a range of 1-10 can only be referenced via its SFI as described in Section 3.1.5.2.

In *UnionPay Integrated Circuit Card Specifications - Basic Specifications*, a data file includes a set of records referenced by record number. The data file referenced in SFI range of 1-10 only includes those data not interpreted by the card, which are not used in the

internal process of the card. The structure of such a file is defined as a linear structure. According to the stipulation in ISO 7816-4, the file structure can be either fixed or linear variable. This is selected by the Issuer and should not affect the read operation of the file according to *UnionPay Integrated Circuit Card Specifications - Basic Specifications*.

3.1.3 Mapping from the File to the File Structure of ISO 7816-4

The following mapping to ISO 7816-4 shall be used:

- A dedicated file (DF) defined in ISO7816-4 is mapped to an ADF or a DDF, to access Elementary files and DFs. The DF located on the top layer of a card is called MF.
- An elementary file (EF) defined in ISO7816-4 corresponds to an AEF. EF will never become the entry of another file.

In *UnionPay Integrated Circuit Card Specifications - Basic Specifications*, if a DF is embedded, the access to the EF connected is transparent.

3.1.4 Directory structure

When there is a payment system environment (PSE) on the card, the IC card must provide a directory structure for the application list from which the Issuer hopes to select in the PSE via directory selection. In this case, all applications are listed in the payment system directory document (DIR file), with DIR file location pointed out by PSE's FCI.

In all cases, the IC card must not use DDF.

The directory structure allows retrieving an application with application identifier (AID).

There must be a code of DIR file existence in any response message to selection of PSE [please refer to SELECT command].

As defined by ISO/IEC7816-5, a DIR file is also an AEF (and also EF), including record structures for the following data objects:

- One or multiple records, with each record containing one or more application templates (with the tag '61') described in this book.
- Other data objects that may appear in directory customized template (with the tag '73'); the data objects included in this template are not defined in *UnionPay Integrated Circuit Card Specifications - Basic Specifications*.

Each record in a payment system directory is a structural data object, with value being one or more directory entries as shown below.

For the format of each record, see Table 1.

Table 1 PSE Directory Record Format

Tag	Data Field Length	Identifier	Directory Entry 1	Directory Entry 1		Identifier	Directory Entry n	Directory Entry n
'70'	(L)	'61'	Length	(ADF)		'61'	Length	(ADF)

Every directory entry in the payment system directory is an application template (tag '61'), it includes Table 2. Besides the data elements in template '73', no other additional data elements can appear in the payment system directory record (tag '70'). Because terminal does not support any unexpected or unexplained Issuer defined operations, template '61' or '73' data elements that appear in payment system directory records should all be ignored.

Any data objects in application templates (tag '61') that are not encapsulated in directory records or other data objects that appear in directory entry but not in Table 2 should all be ignored.

Table 2 ADF Directory Entry Format

Tag	Length	Value		Existence Condition
'4F'	5-16	ADF name (AID)		M
'50'	1-16	Application label		M
'9F12'	1-16	Application preferred name		O
'87'	1	Application priority identifier		O
'73'	Var. Length	Directory customized template		O ¹
	'XXXX' (Identifiers established according to Appendix B UnionPay Integrated Circuit Card Specification. Part 3)	Var. Length	Application supplier, distributor, or IC card supplier, or one or more additional (special purpose) data elements added to the UnionPay Integrated Circuit Card Specification defined template '73' specific tag	O

¹ Other data objects with no relation to *UnionPay Integrated Circuit Card Specifications - Basic Specifications* can also be used to create this data object.

Table 3 Application priority identifier format

b8	b7-b5	b4-b1	Definition
1			Requires cardholder confirmation to choose application
0			Does not require cardholder confirmation to choose application
	'XXX'		Reserved
		0 0 0 0	No priority designated
		'XXXX' (0 0 0 0 excluded)	Application listing or selection order, from 1-15, with 1 having the highest priority

The directory in an IC card is optional, but there is no limit on the number of directories in the card.

3.1.5 File Reference

A file can be referenced via the file name or SFI, according to the file type.

3.1.5.1 Reference via File Name

Any ADF in the card can be referenced via the DF name. The DF name corresponds to its AID; the AID is also used as the beginning of DF name. Every DF name in the card must be unique.

3.1.5.2 Reference via SFI

SFI is used to select AEF. In a given application, any AEF can be referenced via SFI. The SFI uses five bits for encoding and the value is within the range of 1~30. SFI encoding is described in the commands.

The structure of SFI is shown in the table below:

Table 4 SFI Structure

Value	Meaning
1~10	Defined in UnionPay Integrated Circuit Card Specification

11~20	Defined by payment system
21~30	Defined by the Issuer

Every SFI must be unique in an application. AEFs referenced via SFI with a range of 11~20 should fall under payment system distributed management.

3.2 Data Element

Define and explain the relevant data elements required for the card and the terminal in the processing of debit/ credit application data exchange, including the name, identifier and function, etc. of the data element. Please refer to Appendix A, *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification* for details.

3.3 DOL

The terminal may need to transmit a list of designated data elements per card request. In order to reduce data processing in the IC card, the list does not require TLV encoding and only contains several V values. The TL (Tag and Length) list that corresponds to V are pre-defined in the IC card, and referred to as the data object list (DOL). When required, the IC card transmits the fixed format DOL to the terminal, and the terminal follows DOL requirement in assembling the requisite data to send back to the card.

A DOL is a list that is composed of connected TL entries. Each entry represents an additional data element unit added to the DOL. The format of each entry includes 1~2 byte tags to indicate the necessary data object, followed by 1 byte part to indicate the byte length of the data object value. Only basic data objects with tags defined in Appendix A of *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification* could be used in the DOL.

The basic format of the DOL is as follows:

Tag1 Data length Tag2 Data length.....Tagn Data length

The DOLs used in *UnionPay Integrated Circuit Card Specifications - Basic Specifications* include:

- Processing options data object list(PDOL) : DOL (tag and length) that terminal reads from the card during the application selection process, included as FCI in the response data to the SELECT command; data corresponding to the list is transmitted to the card in the GPO command. The data obtained by the card is mainly used for application initialization.
- Card risk management data object list 1 (CDOL1): DOL (tag and length) that terminal reads from the card in the process of reading application data; data corresponding to the list is transmitted to the card with the first Generate AC command. The data obtained by the card is mainly used for card action analysis procedures.

- Card risk management data object list 2 (CDOL2): DOL (tag and length) that terminal reads from the card in the process of reading application data; data corresponding to the list is transmitted to the card with the second Generate AC command. The data obtained by the card is mainly used in transaction termination procedures.
- Transaction certificate data object list (TDOL): DOL (tag and length) that terminal reads from the card in the process of reading application data; data corresponding to the list is transmitted to the card with the first Generate AC command. The data obtained by the card is mainly used for generating the transaction certificate (TC) hash calculation.
- Dynamic offline data authentication object list (DDOL): DOL (tag and length) that terminal reads from the card in the process of reading application data; data corresponding to the list is transmitted to the card with the INTERNAL AUTHENTICATE command. The data obtained by the card is mainly used for dynamic data authentication.

The tags included in the DOL above are referenced from *UnionPay Integrated Circuit Card Specifications - Auxiliary Specifications Part1: Personalization Template Based on Debit/Credit Application*.

The terminal must complete the following steps to generate DOL designated data object value:

1 Read DOL from an IC card.

2 Follow these rules to concatenate all data object values listed in DOL:

a) When the tag of the data object specified in DOL cannot be identified by the terminal or the tag represents a structural data object, the terminal will provide a data element with a length specified by DOL and all numeric parts of the data element must be set as hexadecimal 0s.

b) When a data object in the list is a known data object to the terminal but its value are not be obtained by the terminal, the part of the command field representing the data object must be filled in with hexadecimal 0s.

c) When the length specified in the DOL item is less than the actual length of the data object, it is required to reduce the actual length of the data object to the length specified in DOL. When the data object is in numeric format (n), byte reduction should be conducted from the far left of the data element. When the data object is in other format, byte reduction should be conducted from the far right of the data element. When the specified length is greater than the actual data length, the actual data should be filled in to the designated length.

- When the data object is in numeric format (n), hexadecimal 0s should be used to pad from the beginning of the data object value.
- When the data object is in compressed numeric format (cn), hexadecimal FFs should be used to pad the end of the data object value.

- When the data object is in other formats, hexadecimal 0s should be used to pad the end of the data object value.

d) When a data object in the list is identified by the terminal but not applicable in the current transaction, the fields representing the data object value will be filled in with hexadecimal 0s.

The concatenating sequence of the data object values in the list should exactly correspond to the order of the corresponding data objects in the DOL.

4 Debit/Credit Transaction Processing Flow

4.1 Overview of Functions

The following functions are used in debit/credit transaction processing. Although some steps in mandatory (M) functions may be optional, the functions tagged as mandatory should still be executed in all transactions. The functions tagged as optional (O) are optional and determined by the parameters of the card or/and the terminal.

4.1.1 Application Selection (Mandatory)

When a card is inserted into a terminal, the terminal determines which applications are supported by both the card and the terminal. There are two application selection methods for the terminal:

- 1) The terminal detects the applications supported by both the terminal and the card, and displays them for the user to select.
- 2) The terminal automatically selects the application of the highest priority in the card as designated by the Issuer in advance.

4.1.2 Application Initialization (Mandatory)

After the terminal has selected an application, it must request the card to read the application data of the application, from which it can learn about the functions the card possesses and what support should be provided to the card. The terminal reads the data indicated by the card and uses the list of supported functions to determine the processes to be executed.

4.1.3 Application Data Reading (Mandatory)

The terminal reads out the data of the card used in the transaction processing by using READ RECORD command and the card provides application file locator (AFL) to mark the file where those data are located and record number in the response to the application initialization. The terminal should store all the read-out data objects that can be identified regardless of mandatory or optional for usage in future transactions. It is unnecessary to store the data objects that the terminal is unable to identify (i.e., the terminal is unable to identify the tags of the data objects), but the records containing such data objects may still be required to participate in offline data authentication as a whole, which depends on AFL encoding.

4.1.4 Offline Data Authentication (Optional)

The terminal determines if the offline static or dynamic data authentication should be used to authenticate a card offline based on whether the card and the terminal support those methods. If the terminal supports offline data authentication function and also detects that the card supports SDA or DDA, the terminal should conduct offline data authentication.

The SDA (static data authentication) verifies whether important application data have been tampered with illegally after card personalization. The terminal verifies the static data of

the card using the Issuer's public key, while the card contains the issuer public key certificate and a digital signature, which includes a hash value obtained through encrypting important application data with the Issuer's private key. When the hash value generated from the actual data matches the hash value recovered from the card, it proves that the card data has not been changed.

DDA (Dynamic data authentication) is mainly used to avoid counterfeit cards. There are two kinds of DDA: DDA (standard dynamic data authentication) and DDA/AC-CDA (combined dynamic data authentication). The terminal requests the card to provide the cryptogram generated from the dynamic transaction data encrypted by the IC card's private key and the dynamic transaction data are the unique data generated by the terminal and the card for the current transaction. The terminal obtains the IC card's public key from the card data to decrypt the dynamic signature. A match between the recovered data and the original data proves that the card is not a counterfeit card copied from a valid card. Combined DDA/AC-CDA/application cryptogram generation combine the dynamic signature generation and application cryptogram generation of the card to ensure that the application cryptogram returned when the card action analysis is from a valid card.

4.1.5 Processing Restriction (Mandatory)

The terminal examines whether the application transaction is allowed to continue through processing restrictions. The examined contents include effective date and the expiration date of application, the application version number and other restriction and control conditions defined by the Issuer. The Issuer can use application usage control to set whether the card can be used domestically or internationally, or if the card can be used for cash, purchase or service.

4.1.6 Cardholder Verification (Optional)

The terminal must be equipped with a cardholder identity verification function. Cardholder identity verification is used to confirm the legality of the cardholder to prevent the lost or stolen card from being used. The terminal determines which verification method is to be used via examining the card verification method list (CVMs) of the card. Possible methods are as follows:

- Offline plaintext PIN verification
- Online PIN verification
- Signature
- Fail CVM processing
- No CVM required
- Offline plaintext PIN verification and signature (on paper)
- Cardholder ID verification

4.1.7 Terminal Risk Management (Mandatory)

The terminal must be equipped with a risk management function but the examination can be optional. The terminal can check the data provided by the terminal and the card for floor limit, transaction frequency, new card, terminal abnormal file, Merchant forced transaction online, and random transaction selected for online processing, etc. to achieve risk management.

4.1.8 Terminal Action Analysis (Mandatory)

The terminal must have a terminal action analysis function. Terminal action analysis determines whether to approve the transaction offline, reject the transaction offline, or send an online authorization for the transaction according to the results of offline data authentication, processing restriction, cardholder verification and terminal risk management as well as the risk management parameters set in the terminal and the card. Card rules should be set in Issuer action codes (IACs) field transmitted from the card to the terminal and terminal rules set in the terminal action codes (TACs). After the transaction processing is determined, the terminal requests application cryptogram from the card. Different application cryptograms correspond to different transaction processing: the transaction certificate (TC) should be interpreted as an approval, authorization request cryptogram (ARQC) as online request and application authentication cryptogram (AAC) as rejection.

4.1.9 Card Action Analysis (Mandatory)

The IC card is able to perform the risk management algorithm defined by the Issuer to avoid fraud. When the card receives an application cryptogram request from the terminal, the card performs risk management check to determine if the transaction processing set by the terminal should be changed and the check may include prior incomplete online transaction, Issuer authentication failure or offline data authentication failure on previous transaction, and count or amount checking limits has been reached, etc. IC card is able to determine that the transaction should continue in the following modes:

- Agree to complete offline
- Online authorization
- Reject the transaction

After the check is completed, the card uses application data and application cryptogram session key stored on the card to generate application cryptogram and returns the cryptogram to the terminal. For an offline approved transaction, TC and the data generating TC are transmitted to the Issuer in a clearing message for future cardholder disputes and chargeback purposes. When a cardholder disputes a transaction, TC can be used as the evidence of the transaction and for verifying if the Merchant or the Acquirer has changed the transaction data.

4.1.10 Online Processing (Optional)

When the card or the terminal determines that online authorization is required for the transaction and the terminal possesses the capability for online processing, the terminal transmits the ARQC message generated by the card to the Issuer for online authorization. The message includes ARQC cryptogram, data for generating ARQC and the indicator showing the offline processing result. In online processing, the Issuer should verify ARQC in card authentication method (CAM) process for authenticating the card. The Issuer can, in its authorization decision, take the CAM result and offline processing result into consideration.

The authorization response information transmitted back to the terminal can include authorization response Cryptogram (ARPC) generated by the Issuer (generated from ARQC, Authorization Response Code and card application cryptogram session key). The response can also include the Issuer's script to update the card after it is issued. For details of the response data that the Issuer could return, please see 4.2.10.4.

If the authorization response includes ARPC and the card supports Issuer authentication, the card performs Issuer authentication by validating the ARPC to check if the response is from the real Issuer (or its agent). In order to reset some relevant security parameters in the card, one must successfully pass the Issuer authentication, which prevents criminals from pirating the security characteristics by simulating online processing and fraudulently approving the transaction to reset the counter and indicator of the card. When the card fails in the Issuer authentication, the card transaction will be transmitted for online authorization until the Issuer authentication succeeds. If the card fails the Issuer authentication, the Issuer may set up the card to reject the transaction.

4.1.11 Transaction Completion (Mandatory)

Unless the transaction is terminated in the above steps because of abnormal processing, the terminal must perform this function to complete the transaction.

The card and the terminal perform the last processing to complete the transaction. A transaction approved by the Issuer may be rejected because of the Issuer authentication result in the card and the parameters written by the Issuer. The card uses transaction processing, Issuer authentication result and the rules written by the Issuer to determine if the counter and indicator of the chip card should be reset or not. The card can generate TC to approve the transaction, and generate AAC to reject the transaction.

When the terminal transmits clearing message after authorization message, TC should be included in the clearing message.

4.1.12 Issuer Script Processing (Optional)

When the Issuer includes the script in authorization response messages, although the terminal may not understand the script, the terminal must transmit the script commands to the IC card. Before using those updates, the card should perform security check to ensure that the script is from a valid Issuer and the script is not modified in transmission. The

commands do not affect the current transaction but will mainly affect the follow-up functions of the card such as application unlocking, card locking and PIN changing, etc.

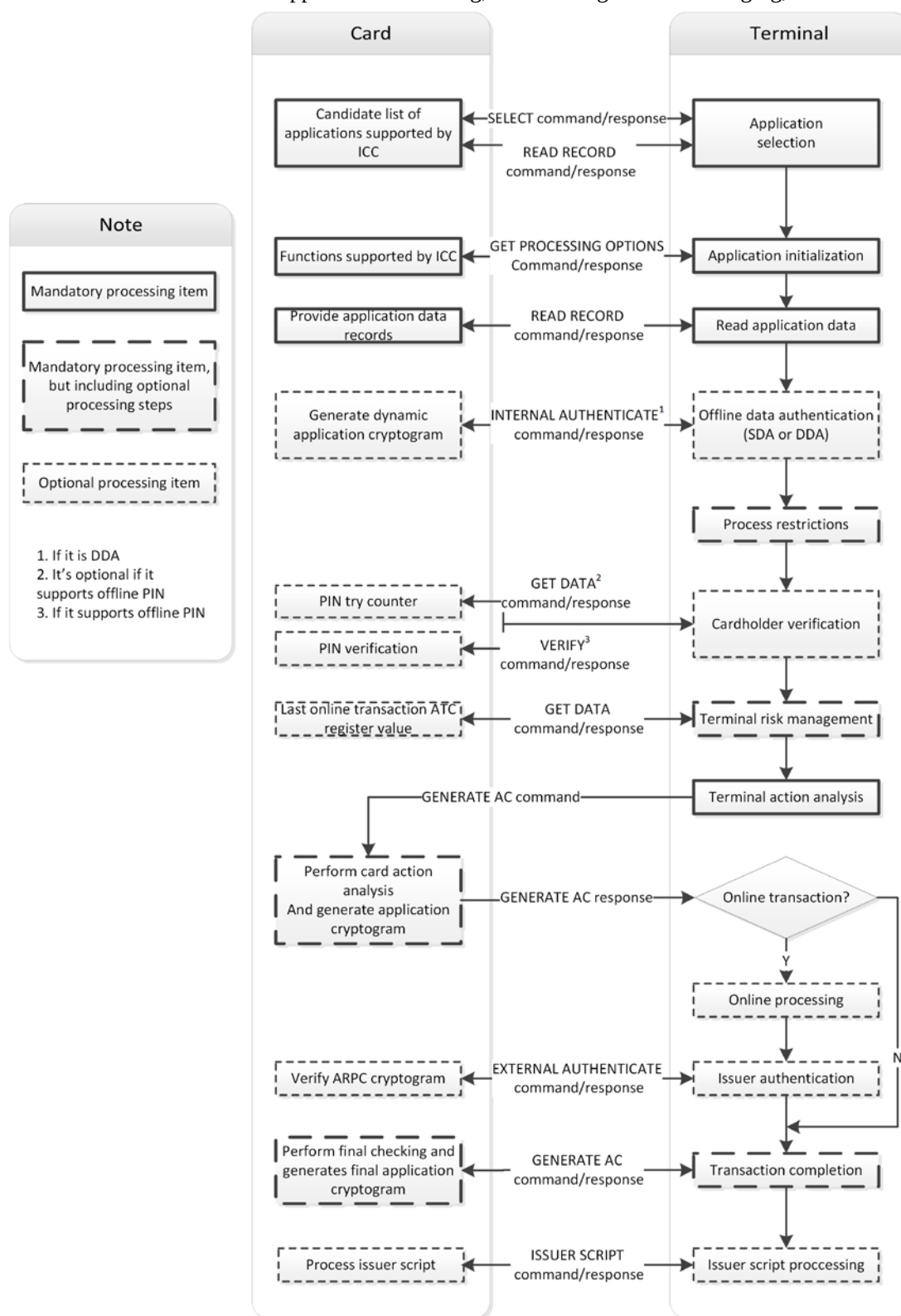


Figure 1 Transaction Process Example

4.2 Transaction Procedures

4.2.1 Application Selection

4.2.1.1 Description

Application selection is a process that determines the application that will be used for conducting the transaction by both the card and terminal. The process is divided into two steps:

- The terminal establishes a list of jointly supported applications for selection.
- Some application in the list is selected and confirmed for processing transaction.

4.2.1.2 Card Data

Table 5 Application Selection - Card Data

Data Element	Description
Application Identifier (AID)	AID consists of Registered Application Provider Identifier (RID) and Proprietary Application Identifier Extension (PIX). It identifies an application described in ISO/IEC 7816-5. If a card contains more than one application with the same AID, then card AID should have a suffix. If there is only one application with this AID, then card AID shouldn't have any suffix unless another application with the same AID might have been added to the card post-personalization.
Payment System Environment (PSE)	PSE is a DDF has name beginning with "1PAY. SYS. DDF01", Directory files associated to the PSE is called the payment system directory.
Directory Definition file (DDF)	DDF is a file that defines the structure of files beneath it, DDF's FCI format is as follows: FCI template <pre> ┌── DF Name │ └── FCI specific template ┌── SFI for DDF SFI │ └── FCI Issuer Defined Data (Optional) </pre>
Application Definition File (ADF)	ADF is an entry to application elementary file (AEF), which contains application data elements. ADF's FCI format is as follows:

	FCI template └─ DF name └─ FCI specific template └─ Application label └─ Application priority indicator (conditional. If card includes many paid accounts, the account that is reflected in the magnetic stripe has priority of 1) └─ PDOL (optional) └─ Preferred language (optional) └─ Issuer code table search (optional. If application preferred name exists) └─ Application preferred name (optional) └─ FCI Issuer defined data (optional), if card supports logging, then log entry (9F4D) data element is in the Issuer defined data
Application Elementary File (AEF)	Application elementary file, including data elements used in application processing.
Directory file	An AEF file, listing the DDF and ADF in the directory. After choosing, use READ RECORD command to access it.
Payment System Directory	DDF relevant directory file with name "1PAY.SYS.DDF01", payment system directory includes entries for ADF and DDF.
File Control Information (FCI)	A response to the SELECT command, depending on different types of documents, response data varies.
Short File Identifier (SFI)	SFI is the pointer to elementary file (EF). 1-10 UnionPay Integrated Circuit Card Specification defined 11-20 payment system defined 21-30 Issuer defined
Application Label	Used for application selection. Exists in ADF's FCI (optional) and ADF's directory entry (mandatory).
Application	Indicates the priority of a given application in a directory and whether

Priority Indicator	cardholder confirmation is needed for selection.
Processing Options Data Object List (PDOL)	In the application initialization step, card needs the terminal data object list, content includes the tag and length of data objects.
Issuer Code Table Search	Code table to be displayed when terminal shows application preferred names in accordance with ISO/IEC 8859.
Application Preferred Name	If the application preferred name exists and terminal supports Issuer code table search entry, at the end of application choices, application preferred name is shown to the cardholder instead of the application label. Application preferred name should be the same as application label, but can also leave it to customer for custom processing.
Short File Indicator (SFI)	SFI is the indicator for application elementary files (AEF).

4.2.1.3 Terminal Data

Table 6 Application Selection --- Terminal Data

Data Element	Description
Application Identifier (AID)	AID consists of registered application provider identifier (RID) and proprietary application identifier extension (PIX).
Application Selection Indicator	Indicating if the related AID in the terminal must strictly match the AID in the card (both the length of AID and content must match) or partially match (first portion of the AID in the card matches the AID in the terminal, the length can be longer). There is only one Application Selection Indicator for each AID in the terminal application list.
List of supported applications	The terminal should save the AID list of the applications supported by the terminal.

4.2.1.4 Command

SELECT

The terminal transmits SELECT command to the card to obtain the relevant information of the applications supported by the card. The information can be Issuer defined parameters such as application selection priority, name of application and the preferred language, etc.

In the response of card to SELECT command, response code is used to indicate processing result. If the response of the card includes processing option data object list (PDOL), PDOL should be processed when the application is initialized.

The response of the card to SELECT command may contain transaction Log Entry and if that data element appears, it indicates that the card supports the transaction log.

READ RECORD

The terminal transmits READ RECORD command to the card to read the record in PSE (if Directory Selection is supported) or other DDFs in the list of AIDs Selection Method. The command includes SFI of the file to be read and the recorded number in the file.

The card responds to READ RECORD command to provide required records for the terminal.

4.2.1.5 Establishing a Candidate Application List

The terminal establishes a list of jointly supported applications in two ways:

- The terminal first tries Directory Selection Method. If the attempt fails, the terminal will adopt AID list method. For Directory Selection Method, the terminal reads a file from the card. The file is a list of applications supported by the card. The terminal adds all applications listed on both card application list and terminal application list onto the candidate list.
- List of AIDs Selection Method is mandatory both to the card and the terminal. For the List of AIDs Selecting Method, the terminal transmits a SELECT command to the card for every application in terminal application list. If the response of the card indicates that the card also supports the application, the terminal will add the application to the candidate list.

4.2.1.6 Identifying and Selecting Application

When there is no jointly supported application, the transaction will be terminated. When there is at least one jointly supported application, the processing flow will be as described in the following sections.

4.2.1.6.1 Terminal Determines Application

If the terminal does not support cardholder application selection or confirmation, the terminal will transmit a SELECT command to the application that does not require confirmation and possesses top priority. If there is more than one application with top priority in the card, the terminal can transmit SELECT command to any one of the applications.

When Directory Selection Method is used to establish application list, the response of SELECT command may indicate that the application has already been blocked. When such a case occurs and there are more available applications in the list of applications, the terminal should transmit SELECT command to the next highest priority application.

4.2.1.6.2 Cardholder Determines Application.

Terminal Supports Cardholder Confirmation

If the terminal does not support cardholder selection from the displayed application list but supports cardholder application confirmation, it will first provide the application with top priority to the cardholder for confirmation. If there is more than one application with the same priority, the terminal can select the first application found or choose one application itself. If the cardholder confirms the selection, the terminal uses SELECT command to select the application.

If the cardholder does not confirm, the terminal will provide application with next highest priority till the cardholder confirms or there is no more available application.

When Directory Selection Method is used to establish application list, the response of the card to SELECT command may indicate that the application has already been blocked. If such case occurs and there are more available applications in application list, the terminal should remove the application from the list of available applications and select the next available application for the cardholder to confirm.

Terminal Supports Cardholder Selection

The terminal supporting cardholder selection provides the cardholder with an application list according to the priority sequence for selection. When there is more than one application with the same priority, the terminal can select one of the applications for processing according to read-out sequence or by itself. The cardholder selects an application from the list and the terminal selects the application with the SELECT command.

When Directory Selection Method is used to establish application list, the response of the card to SELECT command may indicate that the application has already been blocked. When such case occurs and there are more available applications in application list, the terminal should display “try again” and the list of available applications excluding the rejected applications.

If the cardholder does not select any application, the terminal ends the transaction.

4.2.1.7 Flow Chart

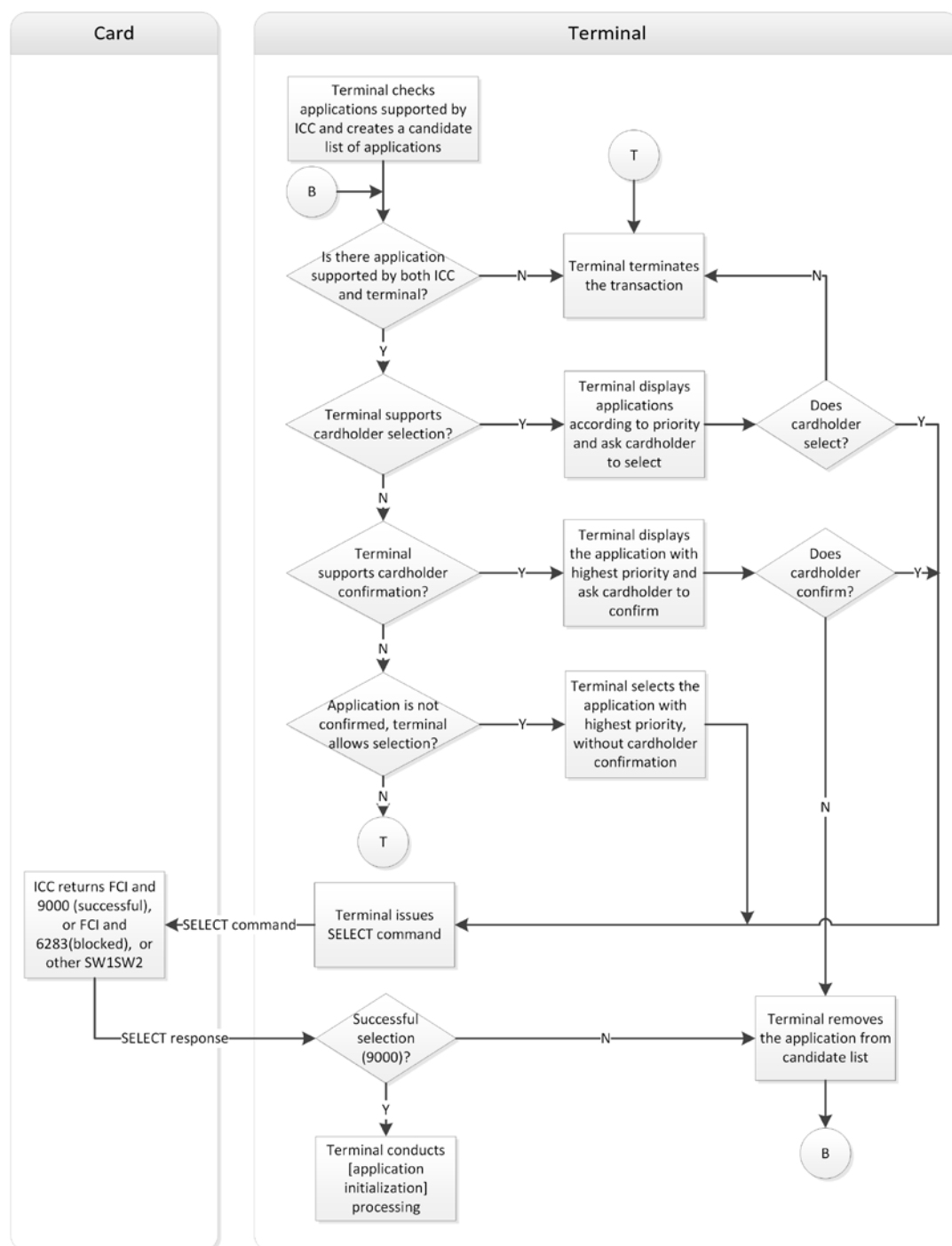


Figure 2 Flow chart of Application Selection Processing

4.2.1.8 Related Post-Processing

Initialization Application Processing

The GET PROCESSING OPTIONS transmitted by the terminal to the card includes all terminal data elements designated by PDOL. When PDOL is supported, PDOL will be included in the SELECT response during the Application Selection.

Reading Transaction Detail Record

For any terminals that need to access the transaction detail records, it can send GET DATA command to obtain the log format data element from the card, and then send READ RECORD command to the card to read the transaction records one by one.

4.2.2 Application Initialization

4.2.2.1 Description

In the application initialization processing, the terminal transmits GET PROCESSING OPTIONS command to the card, indicating the start of transaction processing. When this command is issued, the terminal provides the card with the data elements requested by processing option data object list (PDOL). PDOL is the list of tags and lengths of data elements, provided by the cards to the terminal during the Application Selection. PDOL is the optional data element.

The card provides application file locator (AFL) in response to GET PROCESSING OPTIONS, and AFL command is a list of files and records the terminal is required to read from the card. The card also provides application interchange profile (AIP), which is a list of functions that should be executed during transaction processing.

4.2.2.2 Card Data

Table 4 Application Initialization Processing - Card Data

Data element	Description
Application file locator (AFL)	Indicating the SFI and record range of the card file where the card data required by the terminal in transaction processing flow are located.
Application interchange profile (AIP)	Indicating the list of special functions supported by the card in the application, including SDA, standard DDA, cardholder verification, issuer authentication and combined DDA/AC.
File control information (FCI)	FCI is the related application information of the card in response to the SELECT command transmitted by the terminal.
Processing option data object list (PDOL)	PDOL is an optional list of tags and lengths of terminal data elements requested by the card, which is a part of the card FCI the terminal obtains in the response to SELECT command. The terminal provides the card with the data elements requested in the list in GET PROCESSING OPTIONS command.

4.2.2.3 Terminal data

The terminal transmits the data elements required by the card via PDOL to the card.

4.2.2.4 Command

GET PROCESSING OPTIONS

Through the GET PROCESSING OPTIONS command, the terminal notifies the card that the transaction processing has begun. The terminal uses a PDOL list (if it exists), which is provided when the card selects application. The terminal provides the card with terminal data designated in PDOL via GET PROCESSING OPTIONS command.

4.2.2.5 Processing Flow

4.2.2.5.1 Terminal Processing (Step 1)

For application initialization, the terminal processes are as follows:

- Getting PDOL (if existent) from FCI in response to the SELECT command.
- Transmitting GET PROCESSING OPTIONS command to the card, and within the command, all data elements requested by PDOL are forwarded to the card.

4.2.2.5.2 Card Processing (Step 2)

Once GET PROCESSING OPTIONS command is received, the card performs the following actions:

- Determining the files or records to be read and locating or establishing AFL.
- The terminal may obtain different AIP returns in the response to GET PROCESSING OPTIONS.

4.2.2.5.3 Terminal Processing (Step 3)

The terminal conducts the following processing in response to GET PROCESSING OPTIONS command:

1. Receives the response of the card to GET PROCESSING OPTIONS command.
2. When the response of the card is that the "usage conditions are not met", the terminal will:
 - a. Delete the application from the list of available applications.
 - b. Return to the application selection.
3. When the card responds with AIP and AFL, the terminal begins to read application data.

4.2.2.6 Flow Chart

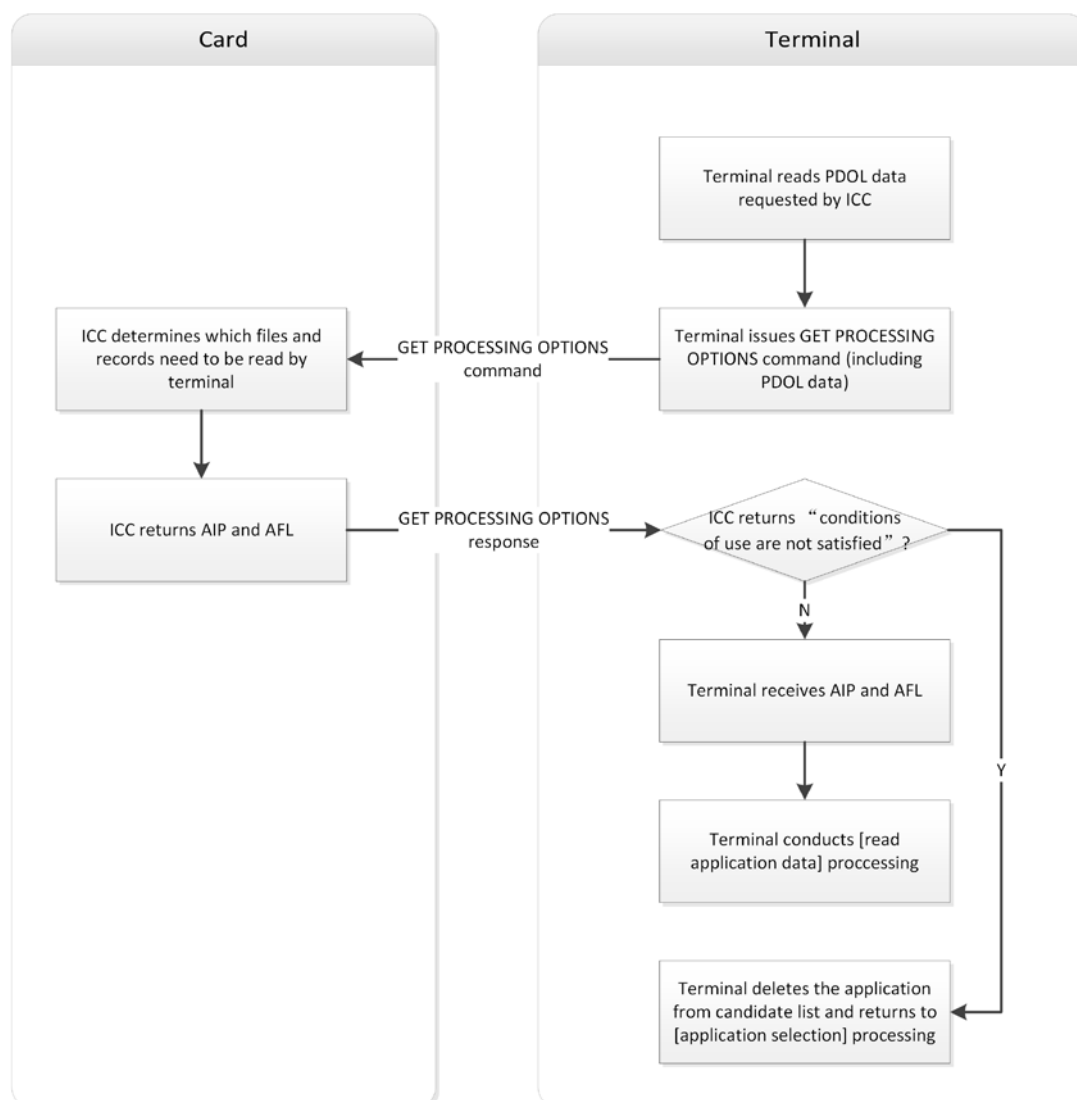


Figure 3 Flow chart of Application Initialization Processing

4.2.2.7 Related Pre-Processing

Application selection

The card provides PDOL (if it exists) as a part of the FCI to the terminal in response to the SELECT command.

4.2.2.8 Related Post-Processing

Reading Application Data

The terminal uses AFL provided by the card in response to GET PROCESSING OPTIONS command to determine which application data should be read from the card and which should be used in offline data authentication.

Offline Data Authentication

The terminal uses AIP provided by the card in response to the GET PROCESSING OPTIONS command to determine if the card supports offline data authentication.

Cardholder Verification

The terminal uses the AIP provided by the card in response to the GET PROCESSING OPTIONS command to determine if the card supports cardholder verification.

Online Processing

The terminal uses the AIP provided by the card in response to the GET PROCESSING OPTIONS command to determine if the card supports Issuer authentication.

4.2.3 Reading Application Data

4.2.3.1 Description

When reading application data, the terminal reads the necessary card data in transaction processing and determines the data used in SDA or DDA.

4.2.3.2 Card Data

Table 8 Reading Application Data – Previously Returned Card Data

Data Element	Description
Application file locator (AFL)	Indicating the file and record range of card data that the terminal will read for transaction processing. Every item designates the initial record and the final record number that will be read from the file and what records will be used in offline data authentication.

Table 9 Reading Application Data - Card Data

Data Element	Description
Application elementary file (AEF)	It is a card data file, including the data used in application processing. It consists of a series of records that are addressed by record numbers. The terminal reads those records with READ RECORD command. The READ RECORD command contains SFI and record numbers, which are obtained from AFL by the terminal, to be read.
Short file identifier (SFI)	SFI is the symbol uniquely identifying application data file. SFI is listed in AFL and the terminal uses SFI to identify the file to be read.

The table below lists the data objects that the IC card must possess when reading records. The other IC card data objects defined in *UnionPay Integrated Circuit Card Specifications - Basic Specifications* are all optional.

Table 10 Reading Application Data - Data Objects the Card Must Possess

Tag	Value	Existence
'5F24'	Expiration date of application	Mandatory
'5A'	Primary account number	Mandatory
'8C'	Card risk management data object list 1	Mandatory
'8D'	Card risk management data object list 2	Mandatory

4.2.3.3 Terminal Data

Terminal data are not used in application data reading function.

4.2.3.4 Command

READ RECORD

The terminal transmits one READ RECORD command to the card for every record to be read. The command includes an SFI of the identified file and a record number to identify the record in the file.

The card provides the requested record in the response to READ RECORD command.

4.2.3.5 Processing Flow

The terminal determines what records should be read from the card according to AFL of the card.

For every AFL item, the terminal requests to read the first designated record via command READ RECORD. When the record returns from the card, the terminal reserves the data object for the subsequent processing. When the AFL item indicates that the record is required for static data authentication during offline data authentication, the terminal will put the record data into static data authentication input list. The terminal will continue to read the file record until the last designated record is read.

4.2.3.6 Flow Chart

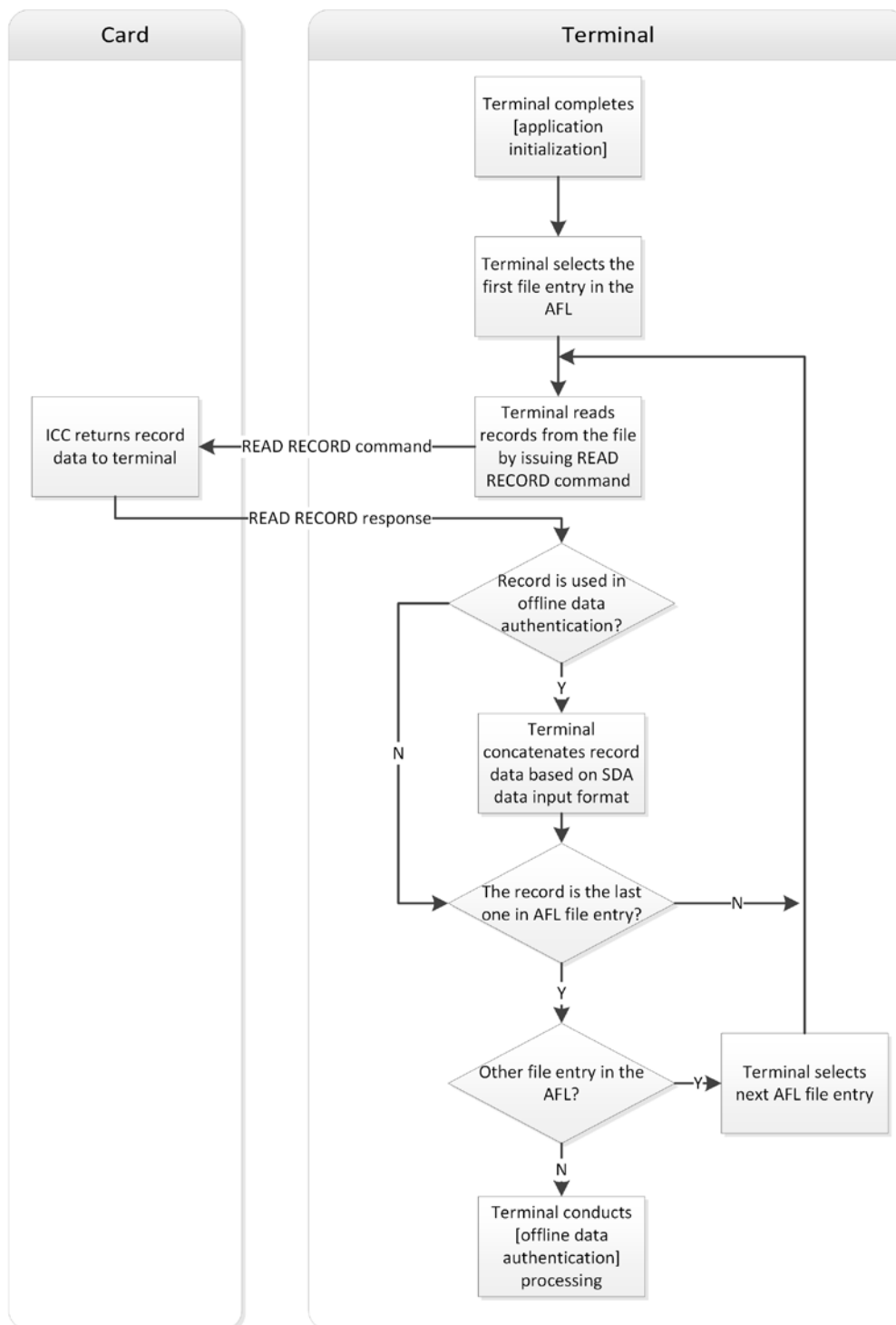


Figure 4 Flow chart of Application Data Read Processing

4.2.3.7 Related Pre-Processing

The terminal uses AFL provided by the card when the application is initialized to read application data.

4.2.3.8 Related Post-Processing

Offline Data Authentication

SDA and DDA verify static data with signature using static data authentication list established when reading application data.

Other Functions

Other functions use the data obtained during reading application data for processing.

4.2.4 Offline Data Authentication

4.2.4.1 Description

Offline data authentication is an authentication process adopted by the terminal for authenticating the data from the card using asymmetric public key technology.

There are two modes for offline data authentication:

- Static Data Authentication (SDA)
- Dynamic data authentication (DDA)

In static data authentication processing, the terminal authenticates the static (not variable) data of the card. Static data authentication ensures that the card data elements selected by the Issuer have not been changed since the card was personalized.

In dynamic data authentication processing, the terminal authenticates not only the static data of the card but also the signature generated from the transaction data which is the only identifier for one transaction the card is able to use. Besides ensuring that the card data elements selected by the Issuer have not been changed since the card was personalized, dynamic data authentication also confirms that the card is a genuine card but not a counterfeit card (reproduced illegally) by copying data from a valid card. The dynamic data authentication can be a standard dynamic data authentication (DDA) or a combined dynamic data authentication/application cryptogram (CDA) generation.

The result of offline data authentication determines if the card and the terminal are going to approve transaction offline, go online for authentication or reject transaction offline. The online authentication system can use the result of offline data authentication in the authentication response decision.

All terminals supporting offline transaction must support static data authentication and dynamic data authentication, with combined dynamic data authentication CDA being optional. For cards that do not support offline transactions, support for offline data authentication is optional. For cards that support offline transactions, dynamic data authentication (DDA) should be supported, but CDA is optional.

4.2.4.2 Key and Authentication

Please refer to the UnionPay Integrated Circuit Card Specifications - Basic Specifications Part4: Debit/Credit Application Security Specification, for detailed information on keys and authentication.

4.2.4.3 Determining the Mode of Offline Data Authentication

Any transaction performs only one method of offline data authentication process. Combined dynamic data authentication/application cryptogram generations enjoy top priority, the secondary is standard dynamic data authentication and the last is static data authentication. Table 11 below indicates that the offline data authentication process to perform should be determined according to the joint support of the card and the terminal.

Table 11 Priority of Offline Data Authentication Processing

Application interchange profile (AIP) of card indicates if the card supports	The terminal supports static data authentication (SDA)	The terminal supports static data authentication (SDA) and standard dynamic data authentication (DDA)	The terminal supports static data authentication (SDA), standard dynamic data authentication (DDA) and combined DDA/application cryptogram generation /application cryptogram generation (CDA)
Static data authentication	Static data authentication	Static data authentication	Static data authentication
Static data authentication and standard dynamic data authentication	Static data authentication	Standard dynamic data authentication	Standard dynamic data authentication
Static data authentication, standard dynamic data authentication and combined DDA/application cryptogram generation	Static data authentication	Standard dynamic data authentication	Combined DDA/application cryptogram generation

4.2.4.4 Static Data Authentication (SDA)

When static data authentication is conducted, the important data of the terminal and the card described in Tables 12 and 13 are used to confirm that the card data have not been changed.

Table 12 Terminal Data Used in Static Data Authentication

Data Element	Description
Public key of certification authority (CA)	Public key of payment system saved in the terminal is used for resuming the public key certificate of the Issuer signed with the private key, issued by certification authority, from the card.
Public key index (PKI) of certification authority (CA)	Used with RID for designating which certification authority public key should be used in offline data authentication.
Registered application provider identifier (RID)	A part of the application identifier for identifying payment system.
Terminal verification results (TVR)	Condition of processing function in view of the terminal.

Table 13 Card Data Used in Static Data Authentication

Data Element	Description
Public key index (PKI) of certification authority	In static data authentication, it is used to identify every public key for offline data authentication and it is also used with registered application provider identifier to identify every authentication public key.
Public key certificate of Issuer	The Issuer public key includes a public key of the Issuer signed with the private key of the certification authority.
Public key exponent of Issuer	The exponent is used in an asymmetric algorithm to recover public key certificate.
Remainder of Issuer public key	When necessary, the remainder of Issuer public key contains the part of public key of the Issuer that has not been listed in the Issuer public key certificate.

Indicator of static data authentication failure	It is an internal indicator, which should be set and saved by the card when static data authentication fails and the transaction is rejected offline.
Signed static application data (SAD)	Static application data is a signature encrypted with the Issuer private key, containing hash value of the important data of the card

4.2.4.4.1 Processing Flow

In static data authentication, the card does not perform any processing. The processing executed by the terminal is briefly described below:

1. Obtaining the public key of certification authority

The terminal uses the public key index (PKI) of the certification authority on the card and the registered application provider identifier to obtain the public key of certification authority and relevant information stored in the terminal.

2. Obtaining issuer public key

The terminal uses the public key of certification authority to recover the issuer public key from the public key certificate of the Issuer. The format of the issuer public key certificate has been verified.

3. Verification of signed static application data

The terminal uses Issuer public key to recover signed static application data, which includes the hash value of the card data calculated via card personalization. The terminal calculates the hash value of actual data elements, which is compared with that of the recovered data. If the hash values are not equal to each other, the data may have been changed and the static data authentication fails.

4. Result of static data authentication

If the above steps are successfully executed, the static data authentication is successful.

If the static data authentication fails, the terminal sets a corresponding indicator in terminal verification result to display the static data authentication result and uses the indicator in the subsequent processing to determine transaction processing.

4.2.4.4.2 Flow Chart of Static Data Authentication Processing

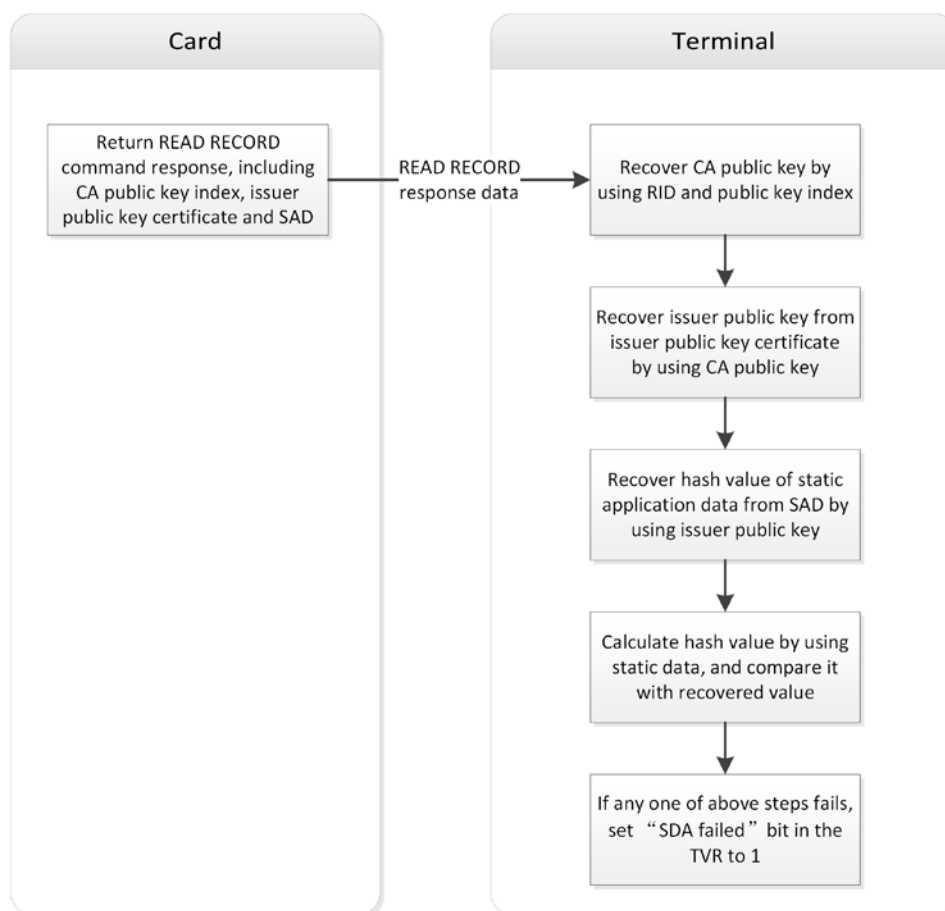


Figure 5 Flow chart of Offline Data Authentication Processing

4.2.4.5 Dynamic Data Authentication (DDA)

When it is required to execute offline dynamic data authentication, the terminal verifies the static data of the card with the issuer public key and the public key of certification authority, the processing flow of which is similar to that of static data authentication. After verifying static data, the terminal applies for dynamic signature from the card. It is required to use INTERNAL AUTHENTICATE command to fulfill standard dynamic data authentication and to use the first GENERATE AC command to fulfill combined dynamic data authentication/application cryptogram generation.

The card uses IC card private key to sign the terminal random number and the dynamic data from the card to generate a digital signature, which is called signed dynamic application data. The signed data generated via combined dynamic data authentication/application cryptogram generation process includes application cryptogram. The card transmits the dynamic signature to the terminal.

The terminal decrypts the signature of the card by using the IC card public key recovered from the public key certificate of the IC card. The recovered data are used in comparison with the actual data to determine if the dynamic data authentication is successful. A

successful dynamic data authentication means that the card data have not been changed and the card is not a counterfeit card.

4.2.4.5.1 Data Element Processed in Dynamic Data Authentication

The terminal will use the terminal data of static data authentication and the data of additional dynamic data authentication described in Table 14 to conduct dynamic data authentication.

Table 14 Terminal Data Used in Dynamic Data Authentication

Data Element	Description
Default dynamic data authentication data object list (default DDOL)	If the card does not provide a data object list for dynamic data authentication, the terminal uses default data object list for dynamic data authentication and the list includes the tag of an unpredictable number to the terminal.
Unpredictable number	The terminal generates an unpredictable number which is to uniquely identify a transaction. The number is transmitted to the card via an INTERNAL AUTHENTICATE command.

All the data for static data authentication, except the signed static application data, are used in dynamic data authentication and in addition, the data described in Table 15 below are also used in dynamic data authentication.

Table 15 Card Data Used in Dynamic Data Authentication

Data Element	Description
Indicator of dynamic data authentication failure	Internal indicator. When standard dynamic authentication fails and the transaction is rejected offline, the internal indicator is set and saved by the card.
Dynamic data authentication data object list (DDOL)	In dynamic data authentication processing, a tag list of the terminal data object should be transmitted to the card.
IC card dynamic data	The card generates a unique number, which is verified by the terminal as a part of the dynamic signature in combined dynamic data authentication/application cryptogram generation.

IC card private key	It is used by the card to generate a dynamic signature.
IC card public key certificate	IC card public key certificate includes an IC card public key signed with the Issuer private key.
IC card public key index	The index is used in asymmetric algorithm to recover signed dynamic application data.
IC card public key remainder	When necessary, the IC card public key remainder includes the part not listed in IC card public key certificate.

All data elements used in standard dynamic data authentication, except dynamic data authentication data object list, are used in combined dynamic data authentication/application cryptogram generation. In addition, the data described in Table 16 are also used.

Table 16 Card Data Used in Combined Dynamic Data Authentication/Application Cryptogram Generation

Data Element	Description
Application cryptogram	Encrypted cryptogram that the card returns in the response to GENERATE AC command. If combined dynamic data authentication/application cryptogram generation is returned with ARQC or TIC, ARQC or TC is a part of dynamic signature authentication.
Cryptogram information data	The card provides cryptogram type information for the terminal verification in combined dynamic data authentication/application cryptogram generation.

4.2.4.5.2 Processing Flow of Standard Dynamic Data Authentication (DDA)

In this processing flow, except for dynamic signature generated by the card, all other steps are executed by the terminal. The processing flow is as follows:

1. Obtaining the public key of certification authority

The terminal obtains the public key of certification authority and the relevant information saved in the terminal by using the public key index (PKI) of certification authority and the registered application provider identifier in the card.

2. Obtaining the issuer public key

The terminal uses the public key of certification authority to recover the Issuer public key from the Issuer public key certificate. The format of Issuer public key certificate has been verified.

3. Obtaining the IC card public key

The terminal decrypts IC card public key certificate including IC card public key and static application data hash value with the Issuer public key. The terminal compares the hash value with that of recovered data for verification. If the hash values are not the same with each other, the dynamic data authentication fails.

4. Generating dynamic signature (standard dynamic data authentication only)

The terminal transmits INTERNAL AUTHENTICATE command including dynamic random number to the card.

Once receiving INTERNAL AUTHENTICATE command, the card encrypts the hash value of the terminal and the card dynamic data with IC card private key to generate a dynamic signature. The card then transmits the dynamic signature to the terminal.

5. Verifying dynamic signature (standard dynamic data authentication only)

The terminal uses the IC card public key recovered from IC card public key certificate to decrypt the dynamic signature. If the hash value of actual dynamic data generated by the terminal is not identical with the recovered hash values, the dynamic data authentication fails.

4.2.4.5.3 Flow Chart of Dynamic Data Authentication Processing

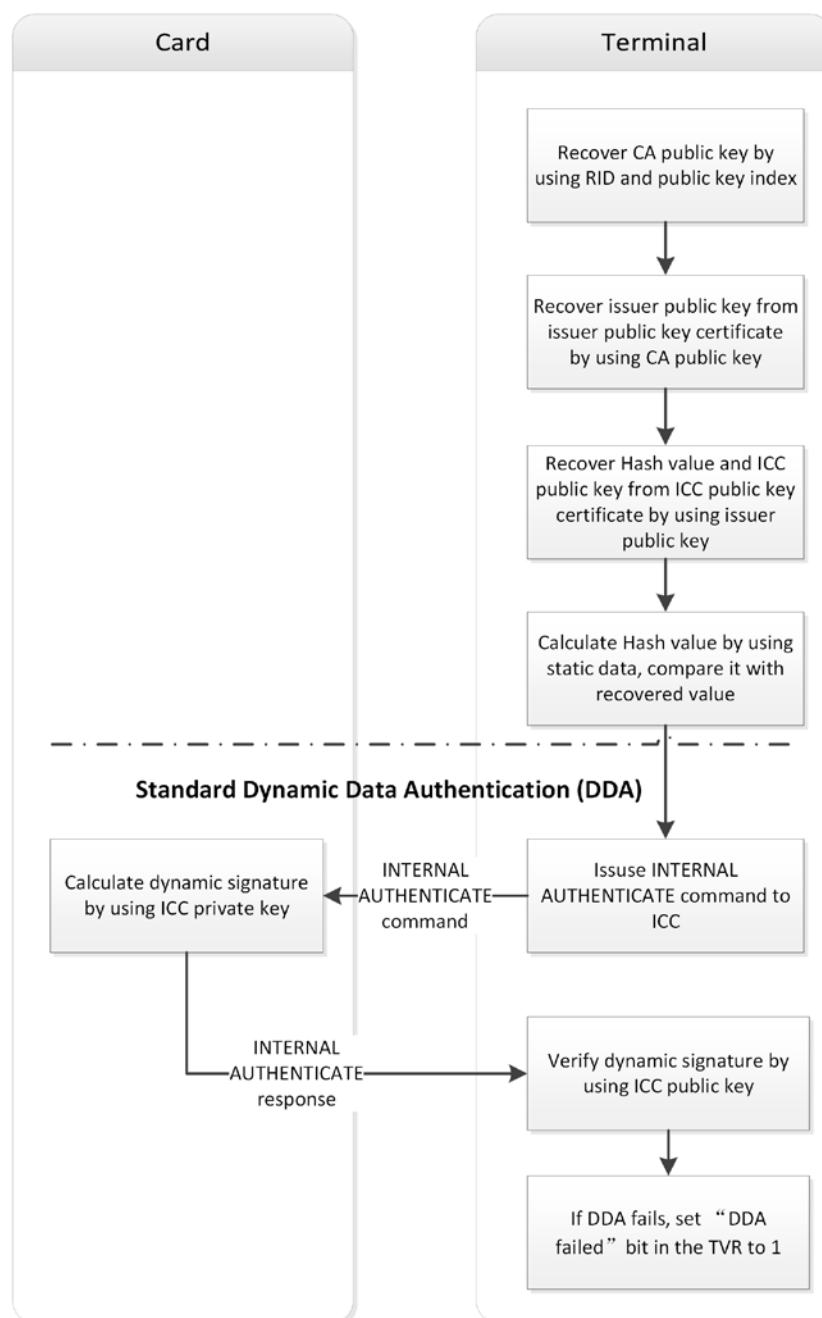


Figure 6 Flow chart of Dynamic Data Authentication Processing

4.2.4.5.4 Processing Flow of Combined Dynamic Data Authentication/Application Cryptogram Generation (CDA)

For combined dynamic data authentication/application cryptogram generation, the terminal performs steps 1-3 for standard dynamic data authentication. It is required for the terminal to use the first GENERATE AC command to generate dynamic cryptogram. No command INTERNAL AUTHENTICATE is used. The requirements and authentication of the cryptogram include the following steps:

1. Generating dynamic signature (combined dynamic data authentication/application cryptogram generation only)

In terminal action analysis, when the terminal requires an online cryptogram (authorization request cryptogram) or offline approved cryptogram (transaction certificate), the first GENERATE AC command indicates that combined dynamic data authentication/application cryptogram generation will be performed. If the application cryptogram is a transaction certificate or authorization request cryptogram, the card will use IC card private key to sign the application cryptogram and the relevant data and returns the dynamic signature to the terminal in response to GENERATE AC command.

2. Verifying dynamic signature (combined dynamic data authentication/application cryptogram generation only)

In card action analysis, if the initial GENERATE AC response contains a transaction certificate or authorization request cryptogram, the terminal uses the IC card public key recovered in step 3 in 4.2.4.5.2 to decrypt the dynamic signature. If the signature is successfully recovered, the processing will go on according to the received cryptogram type. If the signature recovery fails, the transaction will be rejected offline.

4.2.4.6 Related Pre-Processing

Reading Application Data

The terminal reads from the card application data which include the data required to support offline data authentication process. The application file locator and static data authentication tag list indicate the data used in static data authentication for authenticating the hash values of static data and the IC card public key certificate used in dynamic data authentication.

4.2.4.7 Related Post-Processing

Terminal Action Analysis

The terminal uses offline data to authenticate results, card and terminal parameters to determine if the transaction should be rejected offline or online authentication should be conducted or it should be approved offline. If it is required to perform combined dynamic data authentication/application cryptogram generation and the transaction is to be transmitted for online or offline approval, the terminal sets a combined dynamic data authentication/application cryptogram generation indicator in GENERATE AC command.

Card Action Analysis

- Static data authentication and standard dynamic data authentication

If the static data authentication of the last transaction fails and the transaction is rejected offline, the card will set a relevant indicator in CVR. If the dynamic data authentication of

the last transaction fails and the transaction is rejected offline, the card will also set a similar indicator in CVR.

If the static data authentication or dynamic data authentication fails and the transaction is to be rejected offline, static data authentication failure indicator or dynamic data authentication failure indicator should be set.

- Combined dynamic data authentication/application cryptogram generation

If that GENERATE AC command from the terminal is received means that combined dynamic data authentication/application cryptogram generation will be performed, the card will return authorization request cryptogram and transaction certificate application cryptogram. The cryptogram is signed with IC card private key.

Online Processing

- Combined dynamic data authentication/application cryptogram generation

If the returned application cryptogram is a dynamic signature, the terminal decrypts the signature with the IC card public key, if the decryption is successful, the terminal will continue the processing according to the application cryptogram and if the decryption fails, the transaction will be rejected offline.

Completion

After online authentication, the card is allowed to reset static data authentication failure indicator or dynamic data authentication failure indicator according to the issuer authentication options and the result.

If static data authentication or dynamic data authentication fails, the online authentication cannot be completed, the transaction will be rejected offline, thus setting the static data authentication failure indicator or dynamic data authentication failure indicator.

Combined Dynamic Data Authentication/Application Cryptogram Generation

If combined dynamic data authentication/application cryptogram generation fails and the returned application cryptogram is an authorization request cryptogram (AROC), then no online transaction will be executed and a second GENERATE AC command will be transmitted by the terminal to the card, to request AAC. If terminal sends the first GENERATE AC command and the returned application cryptogram is a transaction certificate (TC), the transaction will be rejected offline but no second GENERATE AC command is required.

4.2.5 Processing Restriction

4.2.5.1 Description

The terminal uses the data elements of the terminal and the card to perform processing restriction function. The terminal must support the relevant check of the application version, effective date and expiration date as well as the condition of the transaction point.

4.2.5.2 Card Data

Card data elements used in processing restriction are listed and described in Table 17. Please refer to Appendix A, UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification for detailed description of those data elements and their usages.

Table 17 Processing Restriction - Card Data

Data Element	Description
Application version number	The data element (tag '9F08') indicates the application version of the card and will be used by the terminal in the check of application version number.
Application usage control (AUC)	AUC is an optional data element, which indicates all restrictions of the Issuer concerning card application in terms of area and allowable services. AUC is used by the terminal in application usage control and check.
Country code of the Issuer	Country code of the Issuer is a data element of UnionPay Integrated Circuit Card Specifications, indicating the country issuing the card, and is used by the terminal in the check of application usage control.
Application Effective Date	Application Effective Date is the date if the application begins to be used.
Expiration date of application	After the expiration date of application, the application is prohibited.

4.2.5.3 Terminal Data

Terminal data elements used in processing restriction are listed and described in Table 18. Please refer to Appendix A, UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification for the detailed descriptions of the data elements and the uses.

Table 18 Processing Restriction - Terminal Data

Data element	Description
Application version number	The data element (tag '9F09') indicates the application version of the terminal and will be used by the terminal in the check of application version number.

Terminal capabilities	It indicates the capabilities of the terminal concerning card data input, cardholder verification and security. It is used by the terminal in application usage control and check.
Country code of terminal	The data element indicates the country where the terminal is located and is used by the terminal in the check of application usage control.
Transaction Date	It is the local date if the transaction occurs provided by the terminal and is used by the terminal in the check of effective date and expiration date of the application.
Transaction Type	This data element indicates the type of financial transaction and is used by the terminal in the check of application usage control.

4.2.5.4 Application Version Number Check

The terminal compares the application version number of the card with that of the terminal to see if the two are the same. If they are different, the terminal indicates the inconsistency of the application versions in the terminal verification result (TVR).

4.2.5.5 Application Usage Control and Check

In the process of application usage control and check, the terminal checks the various aspects of the transaction to determine if the processing should continue. The checks are similar to that of service code for transaction performance of magnetic stripe cards and the following transactions include such restriction checks:

- Domestic
 - ✓ Cash
 - ✓ Commodity
 - ✓ Service
- International
 - ✓ Cash
 - ✓ Commodity
 - ✓ Service
- ATM
- Device except ATM

4.2.5.6 Check of Application Effective Date

The check of application effective date confirms that the application has already been effective by verifying the application effective date of the card (if any), which should be earlier than or equal to the current transaction date of the terminal. If the effective date is later than the transaction date, the terminal will indicate in the terminal verification result that the application is not yet effective.

For a card, the application effective date is optional. But for a terminal, the check of application effective date is mandatory if the card's effective date exists.

4.2.5.7 Check of Application expiration date

The check of application expiration date verifies that the application has not expired by confirming that the application expiration date is later than or equal to the current transaction date of the terminal. If the expiration date is earlier than the transaction date, the terminal will indicate in the terminal verification result that the application has expired.

4.2.5.8 Flow Chart of Processing Restriction

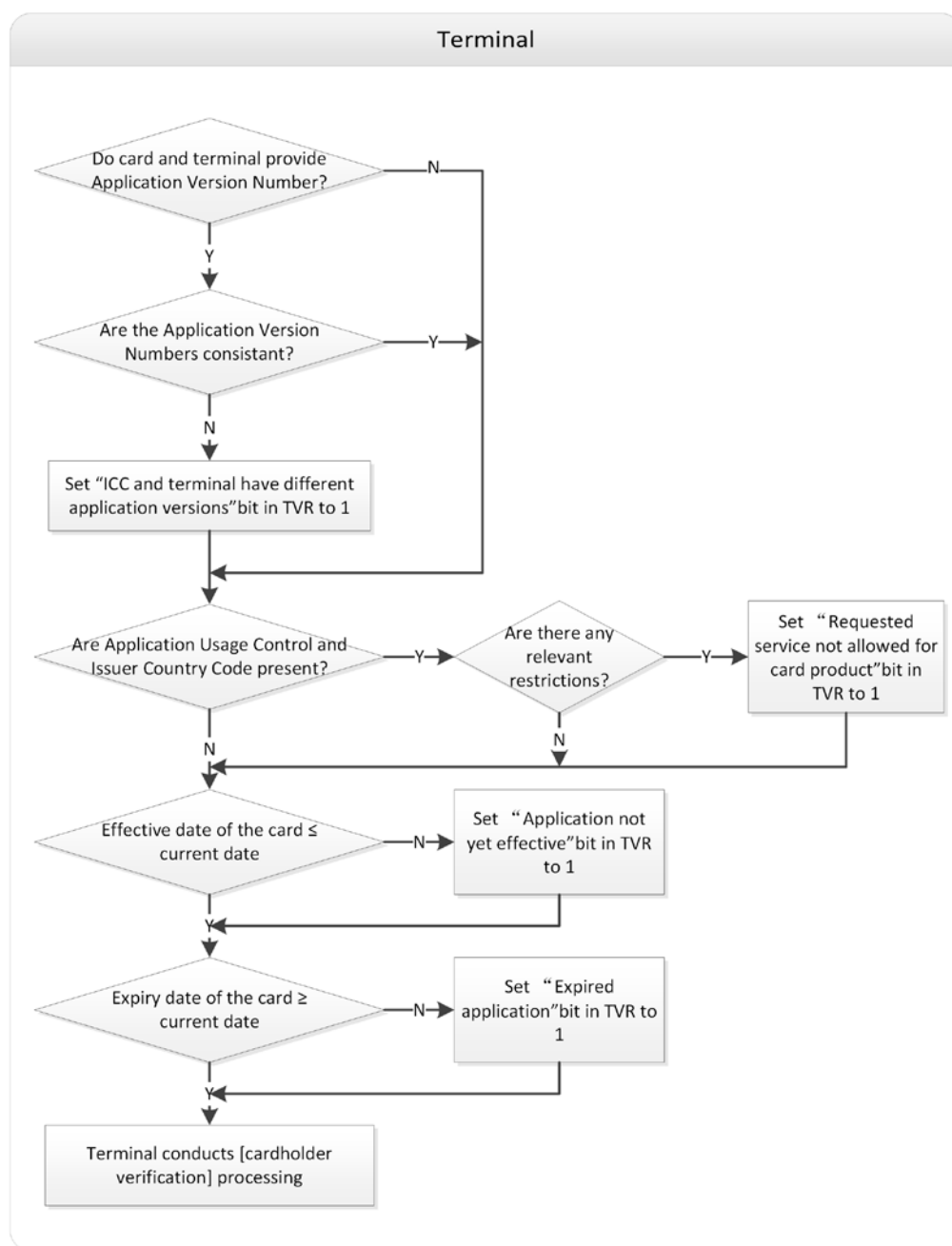


Figure 7 Flow chart of Processing Restriction

4.2.5.9 Related Pre-Processing

Reading Application Data

The terminal obtains application version number and the application expiration date of the card by using the READ RECORD command. If there is application usage control, country code of Issuer, and application effective date, they will also be read from the card.

4.2.5.10 Related Post-Processing

Terminal Action Analysis

In a terminal action analysis, the terminal checks Issuer and terminal action code and determines the kind of processing it must adopt if the application versions are inconsistent and when the card is ineffective, expired, or does not support requested service.

4.2.6 Cardholder Verification

4.2.6.1 Description

Cardholder verification ensures that the cardholder is legal and the card is not a lost or stolen card.

In cardholder verification, the terminal selects a cardholder verification method (CVM) and performs the processing. The result of the cardholder verification method processing will play its role in the subsequent processing.

The supported cardholder verification methods are:

- Offline plaintext PIN verification
- Online PIN verification
- Signature
- Fail CVM processing
- CVM not required
- Offline plaintext PIN verification and signature
- Cardholder ID verification

Signature and ID verification can be combined with offline PIN verification mode. Cardholder verification method processing is designed to support additional cardholder verification, such as the adopted biological identification technology. Confirmation of PIN is completed in the card via offline PIN mode. The offline PIN verification result is included in online authorization message and it must be taken into consideration in authorization decision of the Issuer.

The terminal selects cardholder verification method to be adopted from the cardholder verification method list stipulated by the card. The selecting rules in cardholder verification method list can include transaction type (cash withdrawal or purchase), transaction amount and capabilities of a terminal. If the cardholder verification fails, the cardholder verification method list will also designate a terminal action.

4.2.6.2 Card Data

The terminal uses the card data described in Table 19 and Table 20 in the cardholder verification method list processing. Please refer to Appendix A, *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification* for detailed descriptions of the card data elements and their usages.

Table 19 Cardholder Verification Method List Processing - Card Data

Data Element	Description	
Application interchange profile (AIP)	It contains an indicator to indicate if the card supports cardholder verification. The indicator must be set to 1.	
Cardholder verification method (CVM) list	The card should adopt the sequence of the cardholder verification method list. The card can contain multiple cardholder verification method lists for different environments such as international transactions and domestic transaction. The cardholder verification method list includes the following parts: Amount X— Amount that may be used in cardholder verification method operation condition Amount Y— The second amount that may be used in cardholder verification method operation condition Cardholder verification method item – The cardholder verification method list may include more than one item and every item contains the following subfields:	
	Subfield and	Description
	Cardholder verification method code:	If the cardholder verification fails, the action to be taken will be designated. The next cardholder verification method can be selected for processing or cardholder verification can be terminated.
	Cardholder verification method type:	The cardholder verification method is to perform, such as offline PIN verification.
	Cardholder verification method condition:	The conditions where cardholder verification is to be used, for example if the terminal supports the cardholder verification method type (offline PIN).

Table 20 Offline PIN Verification Processing - Card Data

Data Element	Description
--------------	-------------

Application default action (ADA)	If the offline PIN retry frequencies exceed the limit, the card uses this data element to determine what action to take next.
Card verification result (CVR)	The indicators that the card sets for the following situations: Offline PIN verification performed. Offline PIN verification fails. PIN Try Limit exceeded. Application blocked by IC card because PIN Try Limit exceed.
Counter for PIN retry times	The remaining offline PIN retry times: once cardholder offline PIN verification fails, the counter of PIN retry times will be decreased by 1. If the PIN input by the cardholder is consistent with the reference PIN stored in the card or the script command of resetting PIN retry counter is successfully performed, the PIN retry counter is reset as the upper limit of PIN retry frequencies.
Upper-limit of PIN retry times	The maximum number of times allowed to continuously input the wrong PIN, designated by the Issuer for a specific application
Reference PIN	PIN of cardholder, saved in a secure position in card.

4.2.6.3 Terminal Data

The terminal data described in the table below are used for cardholder verification processing. Please refer to Appendix A, UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification for detailed descriptions of those card data elements and the usages.

Table 21 Cardholder Verification Processing - Terminal Data

Data Element	Description
Encrypting personal identification number (PIN) data	The transaction PIN is encrypted on PIN pad for online authentication.
Personal identification number (PIN) pad secret key	The secret key used on PIN pad for encrypting the input offline PIN and the card reader uses it for decrypting the encrypted PIN. If the PIN pad and the card reader have not been integrated into an integrated device that cannot be interfered from outside, the

	key is a must. The key is different from that for offline encryption PIN.
Terminal capabilities	It indicates the cardholder verification method supported by the terminal.
Terminal verification result (TVR)	Indicators are set for the following situations in terminal verification result: • Cardholder verification was not successful. • Unrecognized CVM. • PIN try limit exceeded. • Request PIN entry, but PIN pad not present or not working. • Request PIN entry, PIN pad present, but no PIN entered. • Online PIN entered.
Transaction Personal Identification Number (PIN)	It includes the data input by the cardholder for PIN verification.

4.2.6.4 Command

The following commands are used for offline PIN processing:

GET DATA

The terminal uses this command to obtain PIN retry counter in order to determine if the PIN input times in the previous transaction exceeds the limit or approaches the limit.

The GET DATA command contains PIN retry counter tag. If the PIN retry counter is in a private data file, the card will return an error response to GET DATA and the terminal will avoid checking PIN input times counter and continue offline PIN verification processing.

VERIFY

This command is used for offline plaintext PIN verification.

The VERIFY command includes the PIN input by the cardholder and initiating the card to compare this PIN with the reference PIN stored in card.

The response of the card indicates one of the following situations:

- The PINs match with each other.
- The PINs are not consistent with each other and the remaining PIN retry time is n. If n equals to 0, the PIN retry times in the current transaction has already exceeded the limit.

- The PIN retry times exceeded the limit in the previous transaction.

If the card and the terminal all support offline PIN processing, they will support the VERIFY command.

4.2.6.5 Processing Flow

Cardholder verification processing is divided into two parts: the cardholder verification method list processing of the card and carrying out cardholder verification.

4.2.6.5.1 Cardholder Verification Method List (CVM) Processing

In cardholder verification method list processing, the card performs no other function except providing the terminal with cardholder verification method list and other necessary data.

The terminal performs the following steps:

1. **Determining if Cardholder Verification Should be Performed** --- If the card supports cardholder verification (just as described in application interchange profile) and reads application data, the card provides a cardholder verification method list and the terminal continues cardholder verification. Otherwise, the terminal will conduct terminal risk management.
2. **Processing Cardholder Verification List Item** --- It should begin with the first item of cardholder verification method list and the terminal performs the following actions:
 - a. Checking if the cardholder verification conditions are met. If the conditions are not met, the terminal conducts the next item of cardholder verification method list.
 - b. If the cardholder verification conditions are met, the terminal will check if the CVM can be recognized or not. If it can be recognized, the terminal will check whether the CVM is supported. If supported, go to Step d. If not supported, the terminal will determine if it is related to PIN verification. If it is PIN verified, the terminal will set the value of "Request PIN entry but PIN pad not present or not working" in TVR to 1, and then go to Step c. If CVM cannot be recognized by the terminal, the terminal will set the value of 'Unrecognized CVM' in TVR to 1, and then go to Step c.
 - c. The terminal checks Bit 7 of CVM code. If it is 1, continue to process next CVM item. If there is no un-processed item in CVM list, the cardholder verification fails. The terminal terminates the cardholder verification.
 - d. Performing the designated CVM. If the cardholder verification is not successful (e.g., offline PIN verification fails), the terminal go to Step c. If the cardholder verification succeeds, the terminal conducts terminal risk management.

3. **If there is no successful cardholder verification while the terminal arrives at the end of the cardholder verification method list, the cardholder verification processing fails**—the terminal sets “cardholder verification failed” bit to 1 in TVR and conducts terminal risk management.

4.2.6.5.2 Flow Chart of the Cardholder Verification Method List (CVM) Processing

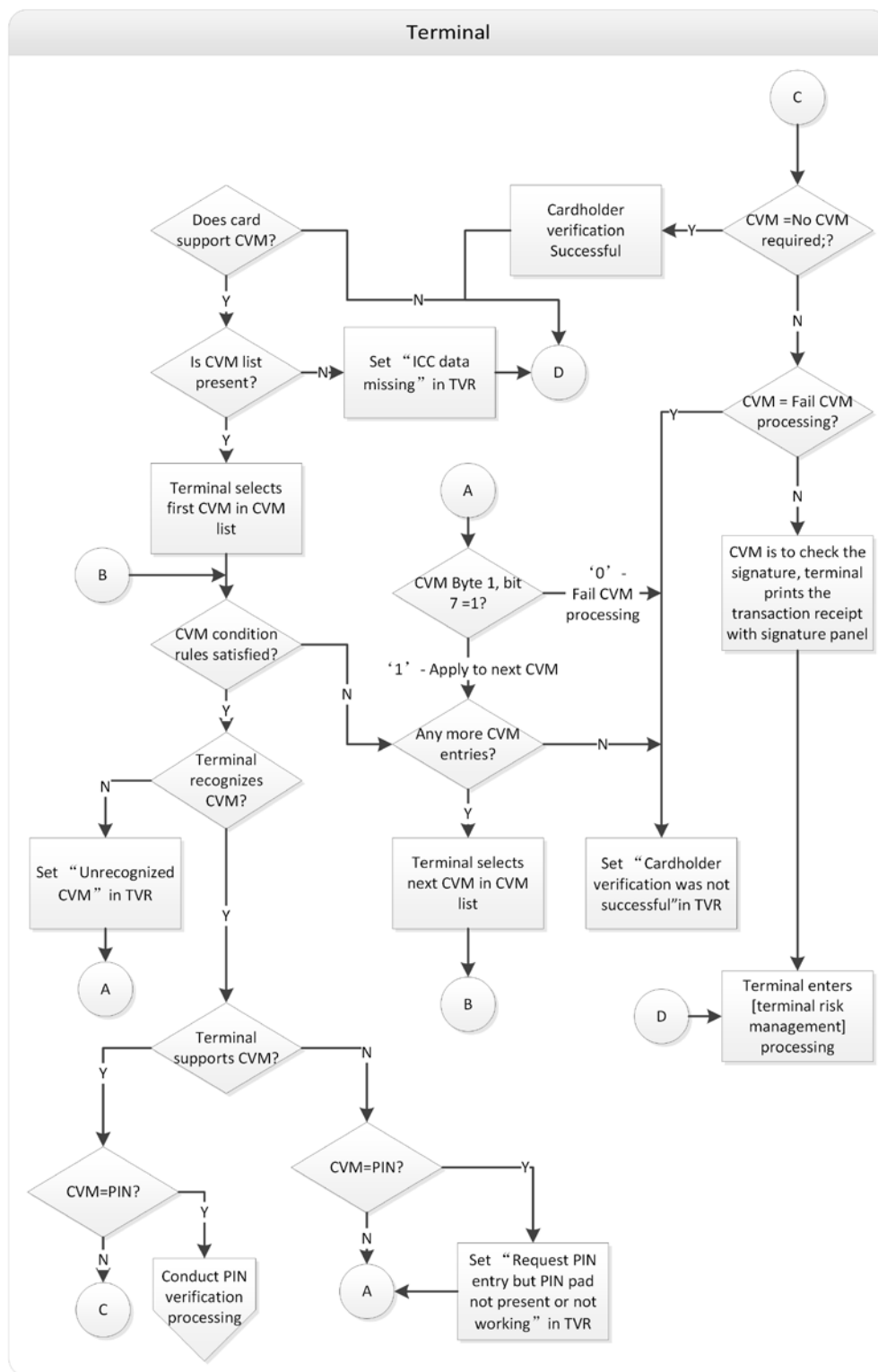


Figure 8 Flow Chart of Cardholder Verification Method List Processing

4.2.6.5.3 Cardholder Verification Processing

4.2.6.5.3.1 Offline Plaintext PIN Verification

In an offline PIN verification processing, the card compares the transaction PIN input to the cardholder with the reference PIN stored in the card for checking. Different from online PIN, offline PIN is not included in the online authorization message. If a transaction is conducted online, the result of the offline PIN verification is included in the online authorization message. The terminal can use the GET DATA command to request for a PIN retry counter from the card, and if the card does not support the transmission of PIN retry counter, the terminal will require consecutive PIN input. If the returned PIN retry counter is zero (there is no remaining PIN retry times), the offline PIN verification fails. If the returned PIN retry counter is 1, the terminal displays *final try*.

If the PIN retry is allowed, the terminal requires the cardholder to input PIN on PIN pad. If the PIN pad and the card reader have not been integrated into an integrated device preventing being interfered from the outside, the PIN is encrypted with a PIN pad secret key and should be decrypted by the card reader. The VERIFY command transmits the transaction PIN input by the cardholder from the card reader to the card in plaintext mode.

The card compares the transaction PIN with the reference PIN stored in card.

- If the two PINs match with each other, the card will return an indicator to display that the offline PIN has been verified and the cardholder verification is completed.
- If the two PINs do not match with each other, the card PIN re-input counter decreases and returns an indicator to display the remaining PIN input times.

If there is no remaining PIN input time, the offline PIN verification fails.

If there are remaining PIN input times, the terminal requests the cardholder to input PIN again and repeat the verification process.

4.2.6.5.3.2 Online PIN Verification

In an online PIN verification processing, the input PIN is encrypted and included in an online authorization message and will be verified by the online system of the Issuer.

The online PIN processing flow is not described in UnionPay Integrated Circuit Card Specifications - Basic Specifications.

4.2.6.5.3.3 Signature

If signature is selected as cardholder verification method, the terminal prints a receipt with a signature panel for the cardholder.

4.2.6.5.3.4 CVM not required

If the cardholder verification method is “CVM not required”, no cardholder verification method shall be performed.

4.2.6.5.3.5 Fail CVM processing

If the cardholder verification fails, it means that the cardholder verification processing failed.

4.2.6.5.3.6 Cardholder ID Verification

The terminal prompts the instruction for the cardholder to present his ID and the type of ID. The ID number obtained from the card will be displayed to the cashier for the cardholder identity comparison and verification.

The two flow charts below describe the PIN verification processing flow.

4.2.6.5.3.7 Flow Chart of PIN Verification Processing (2-1)

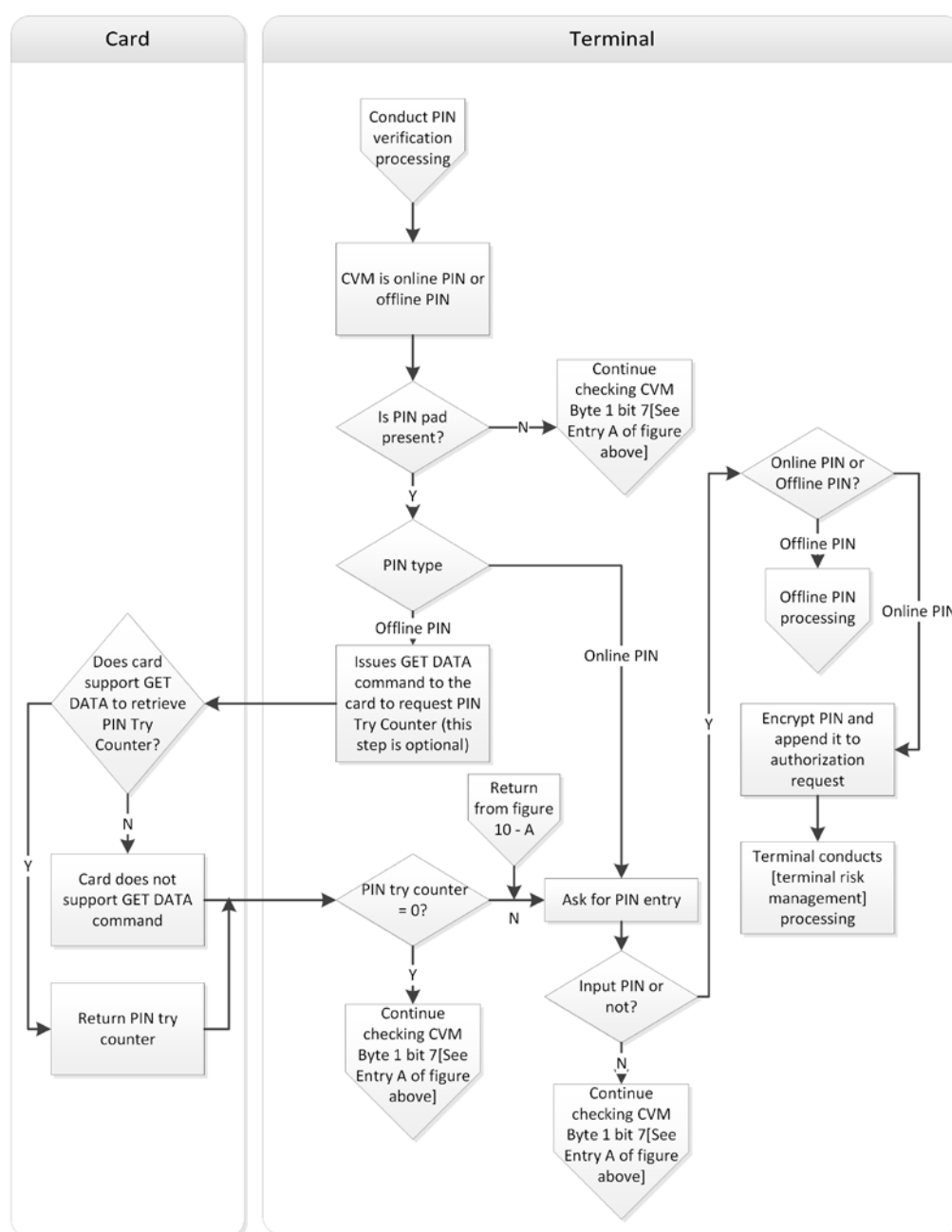


Figure 9 Flow Chart of PIN Verification Processing (1)

4.2.6.5.3.8 Flow Chart of PIN Verification Processing (2—2)

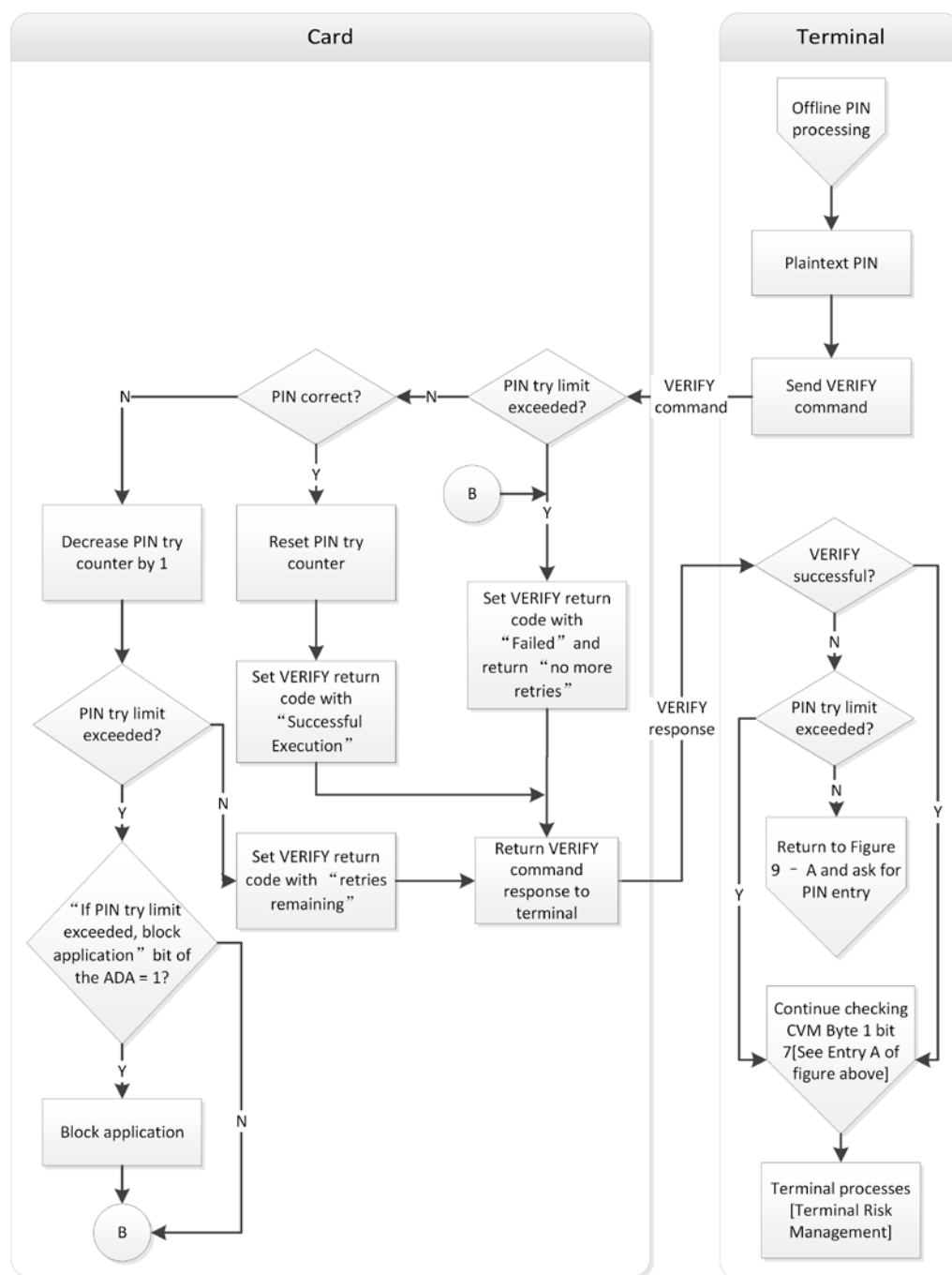


Figure 10 Flow Chart of PIN Verification Processing (2)

4.2.6.6 Related Pre-Processing

Initialization Application Processing

AIP is obtained from the card to indicate if the card supports cardholder verification.

Reading Application Data

The terminal reads the cardholder verification method list and other data used in the cardholder verification processing.

4.2.6.7 Related Post-Processing

Terminal action Analysis

The terminal uses cardholder verification result, the card and terminal parameters called as issuer action code, and the terminal action code respectively to determine if the transaction should be rejected offline or approved offline or transmit an online an authorization request.

Card Action Analysis

If the PIN try frequencies exceed the limit, the card uses the cardholder verification result and parameters in application default action to determine if the transaction should be rejected or conduct an online authorization request.

Online Processing

The authorization request message contains a cardholder verification result which includes an offline PIN check result. Those results should be taken into consideration in the authorization decision of the Issuer. Online authorization message does not include offline PIN.

Transaction Completion

After an online authorization attempt fails, the card uses the cardholder verification result and the parameters of application default action to determine if the transaction should be rejected.

From Issuer to Card Script Command Processing

The PIN CHANGE/UNBLOCK Command can be used to reset PIN retry counter to equal the upper limit of the PIN retry frequency and change the reference PIN.

The APPLICATION UNBLOCK Command can be used to unblock the applications blocked in the cardholder verification processing.

4.2.7 Terminal Risk Management

4.2.7.1 Description

Terminal risk management requires online authorization for large amount transactions and ensures that the chip transaction is able to go online periodically to prevent risks unable to be discovered in the offline environment.

Although the Issuer is forcibly required to set terminal risk management position in AIP as 1 in order to trigger terminal risk management, the terminal should perform terminal risk management without taking the card setting into consideration.

4.2.7.2 Card Data

Card data elements used in terminal risk management are listed and described in Table 22. Please refer to Appendix A of *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part II Debit Credit Application Card Specification* for the detailed descriptions of those card data elements and their usages.

Table 22 Terminal Risk Management - Card Data

Data Element	Description
Primary account number (PAN)	Valid cardholder account number used when terminal exception file checking is conducted
Application transaction counter (ATC)	Quantity of transactions processed since the card is personalized, will be used in the terminal frequency checking.
Register for last online transaction counter	Transaction counter value of last online operation. If the card requires the terminal for terminal frequency check or new card check, this data element and the data elements listed below must all be provided.
Lower limit of consecutive offline transaction	If the terminal can be online, the data element (tag '9F14') is the allowed maximum consecutive offline transaction quantity before the transactions must be operated online defined by the Issuer. This data element is used for terminal frequency checking.
Upper limit of consecutive offline transaction	Data element (tag '9F23') is the allowed maximum consecutive offline transaction quantity before the offline transaction must be rejected defined by the Issuer. This data element is used for terminal frequency checking.

4.2.7.3 Terminal Data

Terminal data elements used in the terminal risk management are listed and described in the table below. Please refer to Appendix A of *Debit/Credit Application Card Specification Part3: Debit/Credit Application Terminal Specification* for detailed descriptions of those card data elements and the usages.

Table 23 Terminal Risk Management - Terminal Data

Data Element	Description
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Amount, Authorized (Numeric)	This data element with numeric value (tag '9F02') stores current transaction amount (excluding adjustment transactions), used for floor limit checking.
Maximum target percentage used for biased random selection	It is used to randomly select transactions for online processing.
Target percentage used for random selection	It is used to randomly select transactions for online processing.
Terminal floor limit	This data element (tag '9F1B') indicates the floor limit of the terminal associated with application identifier. It is used to check the floor limit and randomly select transactions for online processing.
Terminal verification result (TVR)	TVR is a series of indicators used to record the terminal offline processing results. They are used to record the results of terminal risk management checking.
Biased random selection threshold value	It is used for random selection for transaction online processing.
Transaction log	In order to prevent split transactions, the terminal may have recorded the log of approved transactions. The log should at least include primary account number of application and transaction amount and optionally contains the sequence number of the primary account number of application as well as transaction date. The quantity of transaction logs requiring storage and maintenance is not within the scope of <i>UnionPay Integrated Circuit Card Specifications - Basic Specifications</i> . If there is such a log, it can be used in the checking of the floor limit of the terminal.
Transaction status information (TSI)	It indicates the functions performed by the terminal and this data element is not provided in online authorization and clearing message. The terminal uses this data element to indicate that terminal risk management has been performed.

4.2.7.4 Command

GET DATA

If the previous terminal has not read the data, the terminal uses GET DATA command to read from the card the counter register and ATC of the last online transaction.

4.2.7.5 Checking of Terminal Exception Files

If the terminal exception file appears, the terminal checks if PAN of the card is listed in the terminal exception file.

If the card number is listed in the terminal exception file, the terminal sets the position of the “card appears on terminal exception file” as 1 in TVR.

4.2.7.6 Merchant Forcing Transaction Online

For a terminal that can go online, a merchant is able to set the terminal so that every transaction can be processed online.

If a merchant forces transaction online, the terminal will set the position of “Merchant forces online transaction” as 1 in TVR.

4.2.7.7 Floor Limit Checking

The floor limit checking exceeds the terminal floor limit to perform online authorization in a transaction.

The terminal compares the authorized amount with the terminal floor limit. If the transaction amount is greater than or equal to the floor limit, the terminal will set the position of “transaction exceeds the floor limit” as “1” in TVR. Even if the floor limit is 0, the terminal must also perform floor limit check and set the position of “transaction exceeds the floor limit” as “1” in TVR.

If the terminal contains a transaction log, the terminal will check if the previous transaction amount of the same card plus the current transaction amount exceeds the floor limit.

4.2.7.8 Random Transaction Selection

Terminals supporting offline and online transactions are able to randomly select transactions for online processing.

If a transaction is randomly selected, the terminal will indicate it in the terminal verification result. Please refer to the EMV Integrated Circuit Card Specifications for Payment Systems, Version 4.3 Book 3 *Application Specification* for the example of such a processing.

4.2.7.9 Terminal Frequency checking

The frequency checking allows the Issuer to require online processing after a preset quantity of consecutive offline transaction. Terminals that are capable of offline processing must support terminal frequency checking. The Issuer can select not to support frequency checking by the terminal.

If the card provides lower limit of consecutive offline transaction (tag '9F14') and the upper limit of consecutive offline transaction (tag '9F23') in reading application data processing, the terminal will perform terminal velocity checking. If none of these data appear in the card, the terminal will skip this processing.

The terminal transmits the GET DATA command to the card to apply for the counter register of the last online transaction and ATC. The card returns those data elements in response to the command.

The terminal compares ATC with the last online ATC register:

- If the ATC minus the last online ATC register is greater than the lower limit value of consecutive offline, the terminal will set the position of "Lower limit of consecutive offline transaction exceeded" as "1" in TVR.
- If the ATC minus the last online ATC register is greater than the upper limit value of consecutive offline, the terminal will set the position of "Upper limit of consecutive offline transaction exceeded" as "1" in TVR.

Note: In the card action analysis, the card can also perform similar frequency checking. The frequency checking of the card will not affect terminal verification result.

4.2.7.10 New Card Checking

In the new card checking conducted by a terminal, if there are the upper limit value of consecutive offline and lower limit value of consecutive offline, the terminal will check the last online ATC register (if it is provided by the card). After the transaction is approved online based on issuer authentication result and card parameters, the register is reset to the ATC value.

The terminal transmits the GET DATA command to the card in order to request the last online ATC register (if the data element does not appear in the terminal). The card uses the last online ATC register as the response to GET DATA command.

The terminal checks the last online ATC register and if the register is 0, the terminal sets the position of the "new card" as "1" in TVR.

Note: In card action analysis, the card can also perform similar new card checking.

4.2.7.11 Flow Chart (2-1)

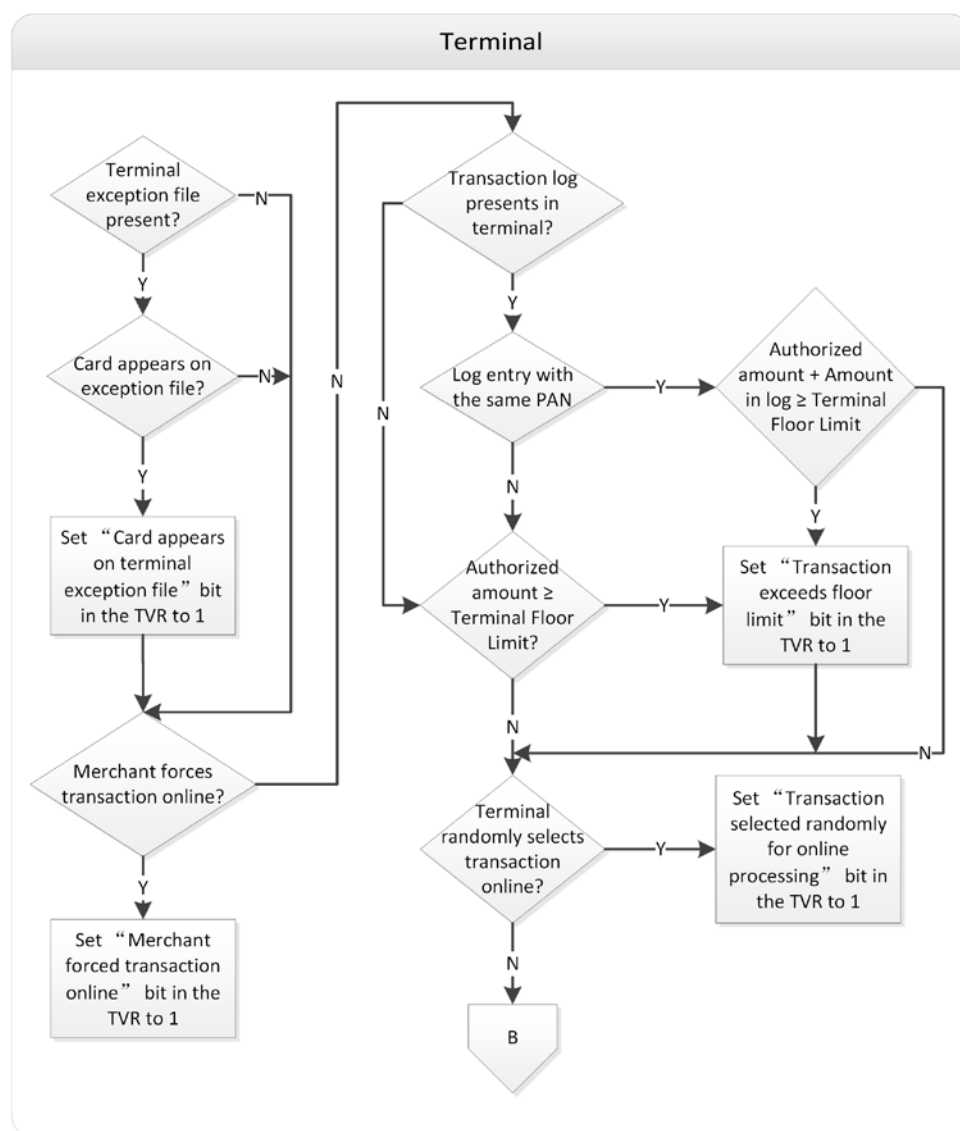


Figure 11 Flow Chart of Terminal Risk Management Processing (1)

4.2.7.12 Flow Chart (2-2)

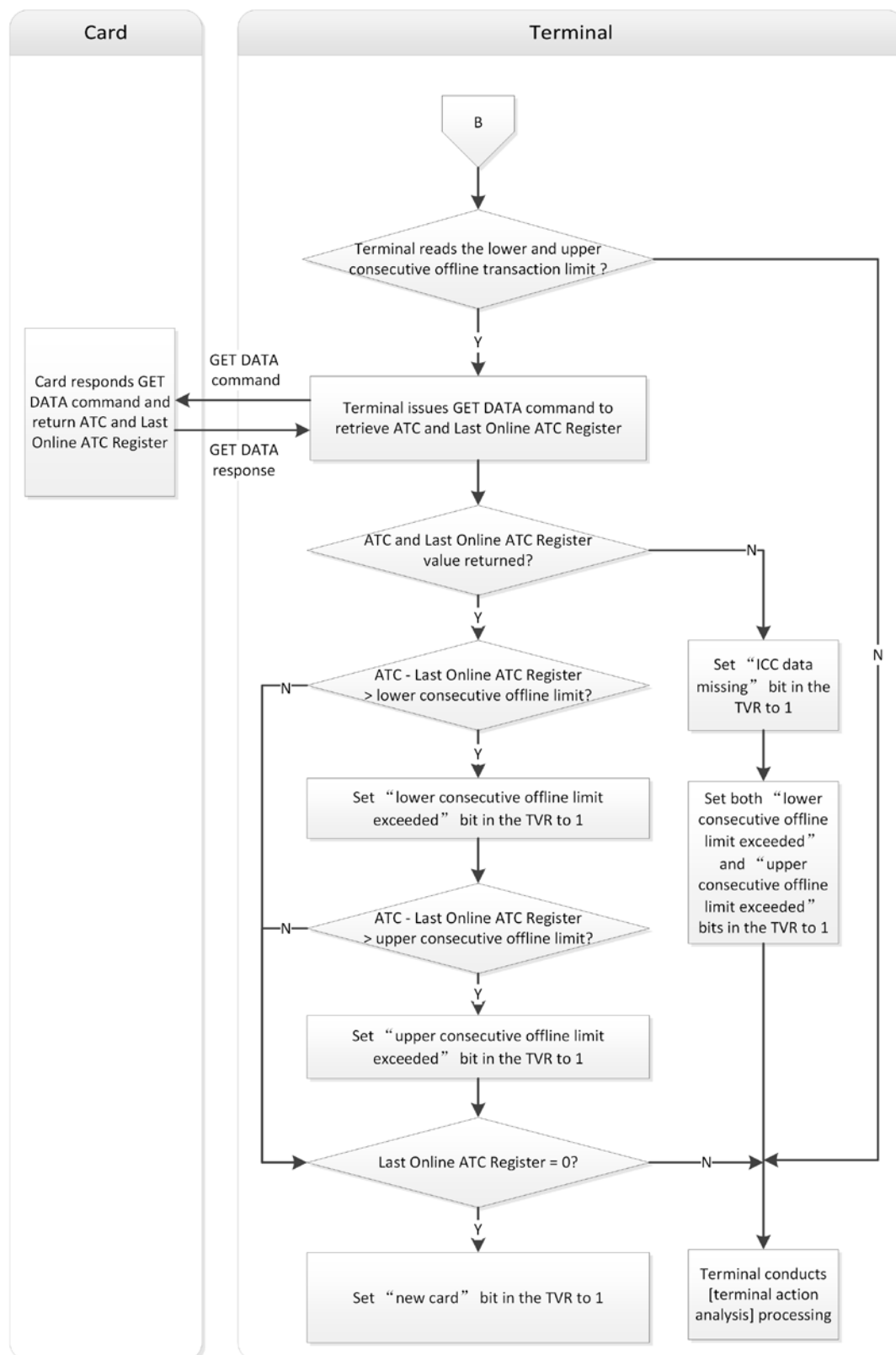


Figure 12 Flow Chart of Terminal Risk Management Processing (2)

4.2.7.13 Related Pre-Processing

Reading Application Data

The following data will be read from the card:

- PAN is used to check terminal exception file.
- If there are upper limit and lower limit of consecutive offline transaction on the card, they are used for terminal frequency checking.

4.2.7.14 Related Post-Processing

Terminal Action Analysis

Based on the following conditions, the terminal determines what actions to take according to the setting of the card and the terminal:

- The card is on terminal exception file
- The merchant forces transactions online
- Exceeding the floor limit
- The transaction is randomly selected for online processing
- Amount or quantity of transactions exceeds the limit in frequency checking
- New card

4.2.8 Terminal Action Analysis

4.2.8.1 Description

In a terminal action analysis, the terminal applies the rules set by the Issuer in the card and by the Acquirer in the terminal in offline processing result to determine if the transaction should be approved offline, rejected offline, or request online authorization.

Terminal action analysis involves two steps:

1. **Checking Offline Processing Result** - the terminal checks the offline processing result recorded by the terminal in terminal verification result to determine if the transaction should be approved offline, rejected offline or online authorization should be requested. The rules defined by the Issuer in the card, such as issuer action code (IACs) and the rules of the terminal, such as terminal action code (TACs) are taken into consideration in the process.
2. **Requesting Cryptogram Processing** - The terminal requires a cryptogram from the card. In terminal action analysis, the decision of offline approval or application for online processing is not final. Based on the result of card action analysis (refer to 6.2.9), the card may override the decision of the terminal, but the decision of offline rejection cannot be overridden.

4.2.8.2 Card Data

The card data elements received from the card previously and used in terminal action analysis are described in Tables 24 and 25. Please refer to *Appendix A*,

UnionPay Integrated Circuit Card Specifications - Basic Specifications Part II Debit Credit Application Card Specification for detailed descriptions of those card data elements and their usages.

Table 24 Checking Offline Processing Result - Card Data

Data Element	Description
Issuer action code (IACs)	Issuer action code has three data elements: Issuer action code - rejection, Issuer action code - online and Issuer action code - default. Every Issuer action code consists of a series of bits corresponding to the bits in TVR. - Issuer action code - rejection bit set to 1 reflects the condition of terminal verification result for the transaction being rejected offline. - Issuer action code - online bit set to 1 shows the online authorization conditions. - Issuer action code - default bit set to 1 is the condition required for offline rejection if online processing does not work. Similar terminal action code (TACs) is defined by the terminal.

Table 25 Requesting Cryptogram Processing - Card Data

Data Element	Description
Card risk management data object list 1 (CDOL1)	Card risk management data object list 1 contains the tag and length of terminal data object and the card uses them to generate the first application cryptogram and conducts other processing.

4.2.8.3 Terminal Data

Please refer to Appendix A, *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification* for detailed descriptions of those terminal data elements and the usages.

Table 26 Check of Offline Processing Result - Terminal Data

Data Element	Description
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Terminal action code (TACs)	Terminal action code has three data elements: terminal action code - denial, terminal action code - online and terminal action code - default. Similar to Issuer action code, terminal action code consists of a series of bits corresponding to the bits in TVR. - Terminal action code - denial bit set to 1 reflects the condition of the terminal verification result for the transaction being rejected offline. - Terminal action code - online bit set to 1 shows online authorization condition. - Terminal action code - default bit set to 1 is the condition required for offline rejection if online processing does not work.
Terminal verification result (TVR)	Terminal verification result is a series of bits set to represent offline processing result during transaction processing period.

Table 27 Requesting Cryptogram Processing - Terminal Data

Data element	Description
Terminal data element	The terminal data elements described in the card risk management data object list 1 are included in the GENERATE APPLICATION CRYPTOGRAM (AC) command.

4.2.8.4 Command

GENERATE AC

The terminal transmits a GENERATE AC command to the card to apply for an application cryptogram. If performing combined dynamic data verification/application cryptogram generation, this command will also appear in the terminal.

This command designates one of the following application cryptogram:

- Transaction certificate (TC) - for approval
- Application authentication cryptogram (AAC) - for rejection
- Authorization request cryptogram (ARQC) - for online

This command also includes the terminal data objects required in card risk management data object list 1.

If a card receives a GENERATE AC command, it conducts card action analysis. The response to this command will not be returned during terminal action analysis.

4.2.8.5 Processing Flow

There are two steps in the terminal action analysis processing:

- Checking offline processing result
- Requesting cryptogram processing

4.2.8.5.1 Checking Offline Processing Result

The terminal checks the result of the offline processing to determine if the transaction should be approved or rejected offline or require an online authorization for the transaction. The rules defined by the Issuer (the issuer action code) in the card and the rules defined by *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part II: Debit/Credit Application Card Specification* (Appendix A: terminal action code) are used in this process. There is an example for the terminal action analysis of *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part II: Debit/Credit Application Card Specification* showing how to use issuer action code (IAC) and terminal action code (TAC) along with terminal verification result (TVR) to determine transaction processing flow.

4.2.8.5.2 Requesting Cryptogram Processing

The second stage of terminal action analysis includes applying for an application cryptogram from the card. The steps of checking offline processing result determine the type of cryptogram to be applied for:

- Offline approval - TC (transaction certificate)
- Conducting online authorization - ARQC (authorization request cryptogram)
- Offline rejection - AAC (application authentication code)

If combined dynamic data verification/application cryptogram generation is performed, this command will also appear in the terminal.

4.2.8.6 Flow Chart

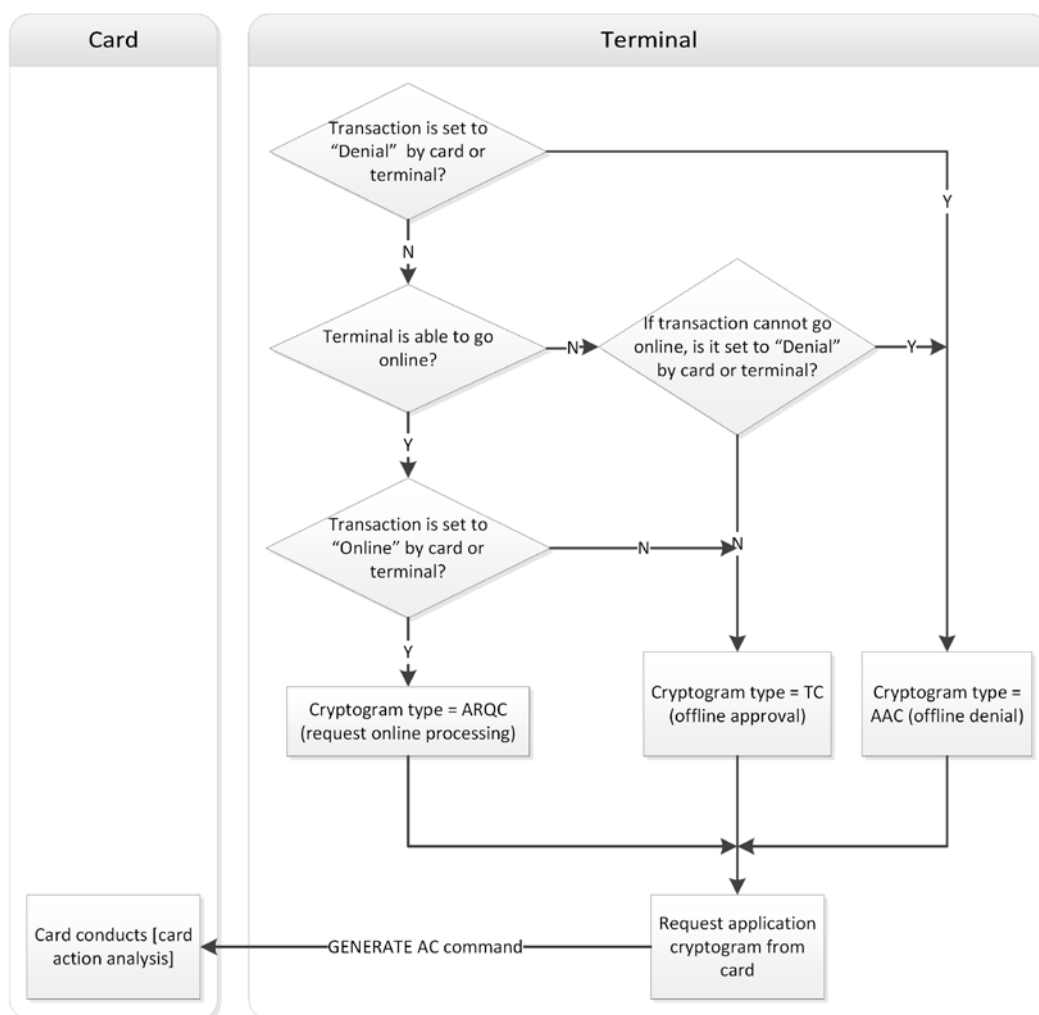


Figure 13 Flow Chart of Terminal Action Analysis Processing

4.2.8.7 Related Pre-Processing

Reading Application Data

The terminal reads application data from the card which include card risk management data object list 1 and issuer action code.

Offline Data Authentication, Processing Restriction, Cardholder Verification and Terminal Risk Management

Bits are set in terminal verification result for those offline functions according to the processing result. In terminal action analysis, the setting of those bits and the issuer action code as well as terminal action code are jointly used to determine transaction processing.

4.2.8.8 Related Post-Processing

Card Action Analysis

In card action analysis, the card performs additional risk management to determine if the offline approval or the decision of requesting online in terminal action analysis should be denied.

4.2.9 Card Action Analysis

4.2.9.1 Description

Card action analysis allows the Issuer to perform frequency checking and other risk management in the card. This section describes the card risk management characteristics specific for UnionPay Integrated Circuit Card Specification, including the following checking:

- Action of last transaction
- New card
- Transaction velocity counter

4.2.9.2 Card Data

The card data elements used in card action analysis are listed and described in Table 28. Please refer to Appendix A of *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification* for the detailed descriptions of those card data elements and their usages.

Table 25 Card Action Analysis - Card Data

Data Element	Description
Application cryptogram	The returned cryptogram that the card responds to the GENERATE AC command: • The application authentication cryptogram returned for rejection is called AAC • Transaction certificate returned for approval is called TC • The authorization request cryptogram returned if online processing is requested is called ARQC
Data required in card risk management data object list (CDOL1)	Data required in card risk management data object list 1. Please refer to <i>Appendix E of UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification</i>, or <i>UnionPay Integrated Circuit Card Specifications - Auxiliary Specifications Part1: Personalization Template Based on Debit/Credit Application</i> for the description of

	CDOL1.
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4.2.9.3 Terminal Data

Terminal data are not used in card action analysis

4.2.9.4 Command

GENERATE AC

The terminal requires the card to return a cryptogram indicating card authorization response result by sending GENERATE APPLICATION CRYPTOGRAM (AC) command to the card for the first time after terminal action analysis is completed. The terminal in this command can also identify if combined dynamic data authentication/application cryptogram (CDA) generation should be performed to generate cryptogram.

4.2.9.5 Processing Flow

After terminal action analysis, the terminal transmits GENERATE AC command to the card and provides the card with the data required in card risk management data object list (CDOL 1) and requests an application cryptogram. Terminal action analysis processing flow is described in *Section 4.2.8*.

The GENERATE AC command received by the card from the terminal contains the type of cryptogram requested by the terminal. The cryptogram type indicates the decision of the terminal for the transaction after the terminal action analysis (approving offline, reflecting offline, or requesting for online authorization).

4.2.9.5.1 Card Risk Management

If the supported and required data by the card can be used, the card performs the following card risk management actions:

- Last transaction action:
 - ✓ The online authorization has not been completed.
 - ✓ When last online transaction was conducted, issuer authentication failed.
 - ✓ The static data authentication of last transaction failed.
 - ✓ The dynamic data authentication of last transaction failed.
 - ✓ The performance situation for the issuer script command of last transaction
 - ✓ PIN retry times of last transaction exceeded the limit.
- New card checking

- Whether the offline processing times of the following items exceeds the limit should be checked in frequency checking:
 - ✓ Total number of consecutive offline transactions;
 - ✓ Total number of international consecutive offline transactions calculated in terms of currency;
 - ✓ Total number of international consecutive offline transactions in terms of country;
 - ✓ Total accumulated amount of all offline transactions with designated currency;
 - ✓ Total offline transaction amount with designated currency and second currency.

4.2.9.5.2 Card Response Decision

The card determines transaction response according to card risk management result. The cryptogram returned from the card can be different from the cryptogram type requested by the terminal:

- The card may apply for online authorization or offline rejection, without taking into consideration the decision that the terminal has approved offline;
- The card may reject the transaction without taking into consideration the decision that the terminal has applied for online authorization.

Table 29 Card Action Analysis - Response of the Card to GENERATE AC Command

		Response of the Card		
		AAC	ARQC	TC
Request of the Terminal	AAC	Rejection	-	-
	ARQC	Rejection	Applying for online processing	-
	TC	Rejection	Applying for online processing	Approval

4.2.9.5.2.1 Standard Response to GENERATE AC

The card uses the data provided by the terminal and the card to generate a cryptogram which is based on symmetric algorithm. The required data are described in detail in *Appendix D, UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2:*

Debit/Credit Application Card Specification. The symmetric key and algorithm required in cryptogram generation process are described in detail in *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part4: Debit/Credit Application Security Specification*.

The card returns the cryptogram to the terminal in the response to GENERATE AC. The cryptogram type in the response indicates the processing decision of the card for the transaction (offline approval, offline rejection or application for online authorization).

4.2.9.5.2.2 The GENERATE AC Response for Combined Dynamic Data Authentication/Application Cryptogram (CDA) Generation

If the terminal indicates in GENERATE AC command that CDA will be performed and the cryptogram type returned by the card in the GENERATE AC response is approval (TC) or for online authorization (ARQC), the card will encrypt the application cryptogram, cryptogram information data and other data with the private key of the IC card. The card will return the signature data to the terminal in the GENERATE AC response.

4.2.9.6 Flow Chart

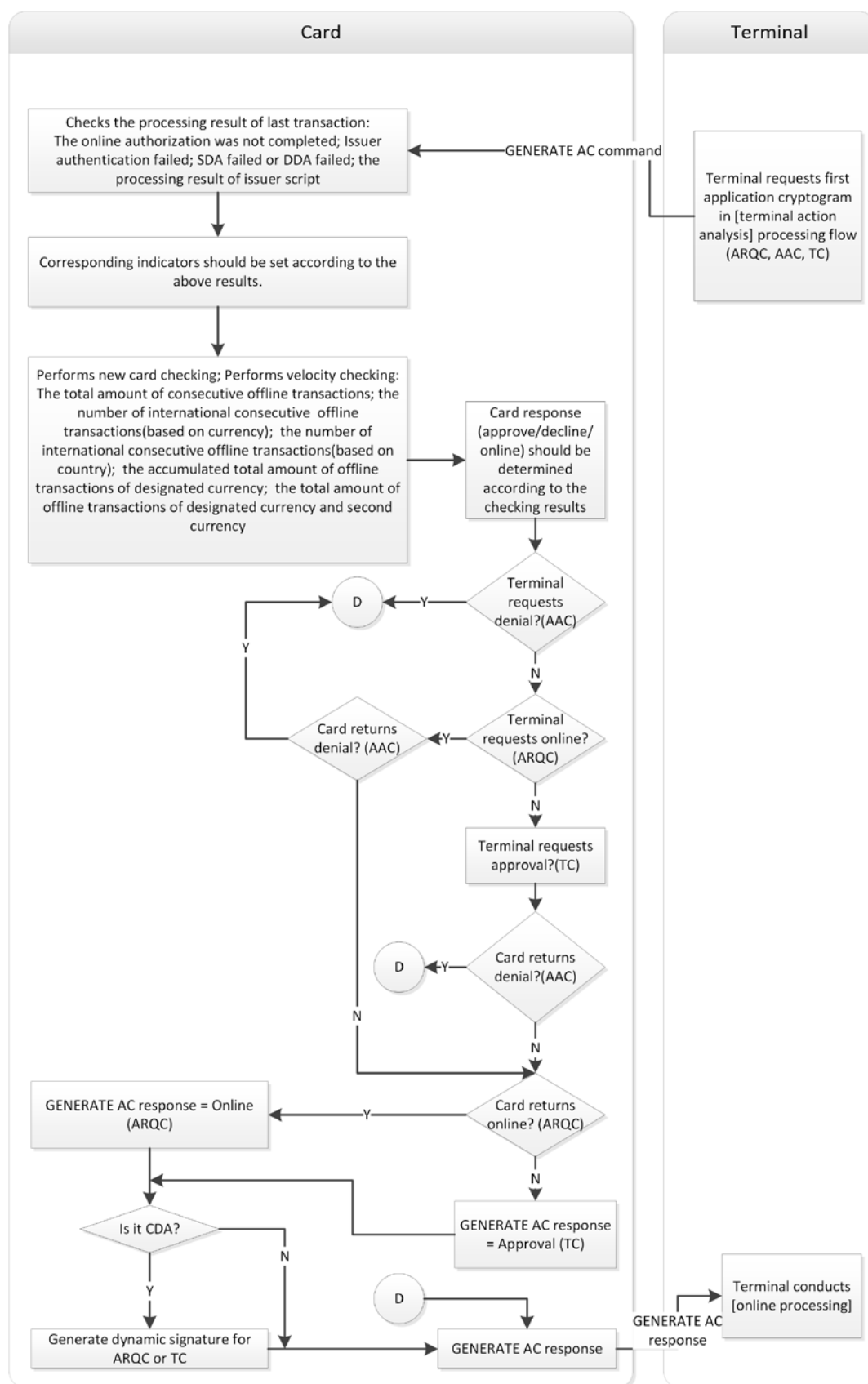


Figure 14 Flow Chart of Card Action Analysis Processing

4.2.9.7 Related Pre-Processing

Reading Application Data

The terminal reads card risk management data object list 1 (CDOL1) from the card.

4.2.9.8 Related Post-Processing

Transaction Completion

If online processing is required, but the terminal is unable to transmit transaction online, the card and the terminal perform other processing to determine if the transaction should be approved or rejected offline.

By using issuer action code (IAC) - rejection and terminal action code (TAC) – denial, the terminal performs other analysis (similar to terminal action analysis) to determine that the cryptogram type (AAC or TC) finally is requested in GENERATE AC command.

The card also performs the following card risk management checking to determine the final transaction processing result:

- Frequency checking of all consecutive offline transactions (upper limit)
- New card
- Offline PIN verification has not been performed.

4.2.10 Online Processing

4.2.10.1 Description

The online processing allows the host of the Issuer to determine if the transaction should be permitted or rejected according to the host risk management parameters set by the Issuer. Compared with traditional online fraud checking and credit checking, the host authorization system should also perform online card authorization by additionally using the dynamic cryptogram generated by the card, also based on the result of offline authentication.

The data returned from the Issuer can include the cryptogram generated by the Issuer and the data for card updating; wherein, the cryptogram generated by the Issuer is used in card authentication for the authenticity of the returned data.

4.2.10.2 Card Data

The card data used by the terminal are as described in Table 30:

Table 30 Online Processing - Card Data Used by the Terminal

Data Element	Description
The Generate AC command	The response data include:

response data	• Cryptogram type (If the transaction requires online authorization, it is authorization request cryptogram ARQC) • Application cryptogram (AC) • Application transaction counter (ATC) • Issuer application data
Application interchange profile (AIP)	The terminal obtains AIP from the card if the application initialization processing is conducted and one of the bits indicates if the card supports issuer authentication.

In the Issuer authentication processing, the data used in the card are as in Table 31:

Table 31 Online Processing - Data Used in the Card

Data Element	Description
Authorization request cryptogram (ARQC)	It is generated by the card in the comparatively earlier steps of the transaction. ARQC and authorization response code will be used as input data in authorization response cryptogram (ARPC) confirmation processing.
Application cryptogram session key (UDK)	It is the DES key used in ARPC confirmation processing, and is the same one as the key used in ARQC generation.
Card verification result (CVR)	If issuer authentication fails, the corresponding bit will be set to 1.
Issuer authentication failure indicator	If issuer authentication fails, the bit will be set to 1.

4.2.10.3 Terminal Data

According to issuer authentication situation, the data elements in the terminal that need to change are as in Table 32:

Table 32 Online Processing - Data that Modified by the Terminal

Data Element	Description
Terminal verification result (TVR)	If issuer authentication fails, the corresponding bit will be set to 1.

Transaction status information (TSI)	After the issuer authentication has been performed, the corresponding bit should be set to 1.
--------------------------------------	---

4.2.10.4 Online Responding Data

The responding data the Issuer may return to the Acquirer, if there is any, are listed in Table 33, the Acquirer must transmit the data to the terminal.

Table 33 Online Processing - Response Data the Issuer May Return

Data Element	Description
Issuer authentication data	It includes the following sub-items: • Authorization response cryptogram (ARPC): Cryptogram generated by the issuer host system. • Authorization response code: Response code used in ARPC generation.
Issuer script	Some command data transmitted by the Issuer to the card and the data will be used for card data updating.

4.2.10.5 Command

EXTERNAL AUTHENTICATE command is used in online processing.

If issuer authentication is performed, the terminal should verify the correctness of authorization response cryptogram (ARPC) via EXTERNAL AUTHENTICATE command using the issuer authentication data requested from the Issuer. It can be known whether the authentication is successful or not through the command response.

4.2.10.6 Processing Flow

Standard online processing includes online request and online response. If necessary, Issuer authentication can be performed. If CDA has been performed, the processing flow should also include verification of dynamic data.

4.2.10.6.1 Online Request

The online request processing differs based on whether CDA has been performed or not.

Execute CDA

If it is indicated in Generate AC command that CDA has been performed and the returned cryptogram is ARQC or TC, the terminal will conduct the following processing:

- The terminal uses the recovered IC card public key to decrypt dynamic cryptogram and obtain application cryptogram.

- If the hash value does not match the value in TVR, the CDA fails, leading to processing completion.
- If the hash values match, standard online processing is conducted.

Standard Online Request Processing

If the card returns ARQC to the terminal with Generate AC command while the terminal being capable of online processing, the terminal sends online authorization message.

If the card does not return ARQC or the terminal is incapable of online processing, then goes to transaction completion step.

4.2.10.6.2 Online Response

After the online request message is successfully sent to the Issuer, the terminal receives the response message returned from the Issuer, in which the Issuer command script or cryptogram used for changing card information can be included or both of them can be included, for confirming that the response message is really returned from the legal Issuer. If the Issuer authentication data is included in online response while the card supports issuer authentication, issuer authentication is performed, otherwise goes to transaction completion step.

4.2.10.6.3 Issuer Authentication

The terminal transmits an EXTERNAL AUTHENTICATE command to the card for performing issuer authentication and the card uses the previously generated ARPC, issuer authorization response code and the Unique DEA Key (UDK) stored in a special safe area of the card to verify the legality of ARPC.

Both the card and the terminal should record the issuer authentication result.

- The card should set issuer authentication result and issuer authentication failure identifier in card verification result (CVR) and returns the result to the terminal in EXTERNAL AUTHENTICATE command response message.
- Before completing the processing, the terminal will set issuer authentication result and transaction status information (TSI) in terminal verification result (TVR).

4.2.10.7 Flow Chart

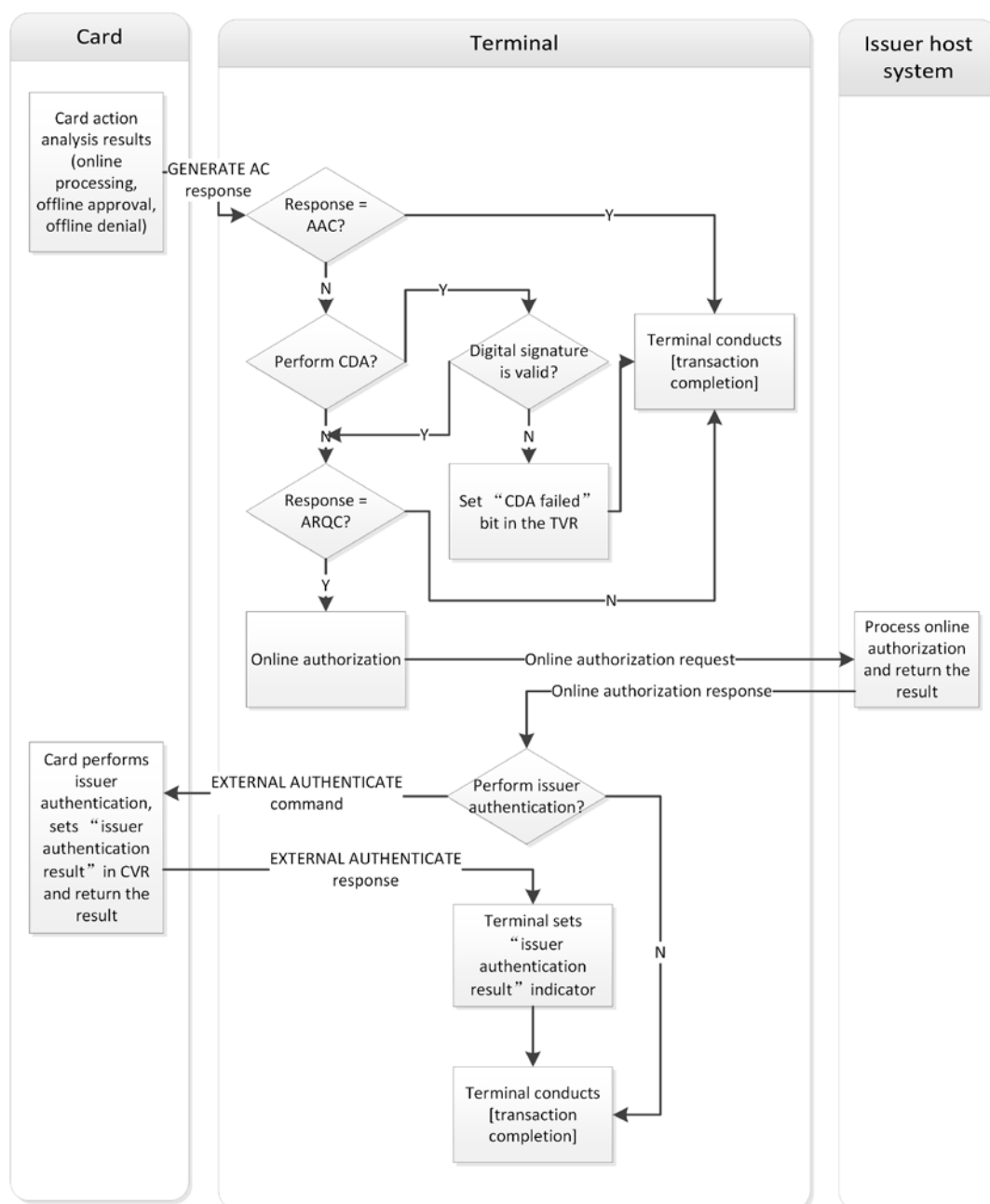


Figure 15 Flow Chart of Online Processing

4.2.10.8 Related Pre-Processing

Card Action Analysis

If online authorization is required after card analysis, the type of cryptogram returned by the card is ARQC.

4.2.10.9 Related Post-Processing

Processing Completion

In processing completion, the card uses issuer authentication result and card parameters to help determine how to process the transaction as well as whether to reset relevant indicators and counters.

Issuer Script Processing

If issuer command scripts are included in online processing scope message, the terminal will need to transmit the command scripts to the card to perform.

4.2.11 Transaction Completion

4.2.11.1 Description

The terminal and the card perform completion to end transaction processing, mainly including the following actions:

- If online processing is required but the terminal does not support online processing or online authorization is unable to be completed, the terminal and the card determine whether the transaction can be completed or rejected offline through other analysis.
- If the terminal fails in performing CDA, the terminal processes according to following methods:
 - ✓ If the card requests ARQC, the terminal will request AAC (rejection cryptogram) when the second Generate AC is generated.
 - ✓ If the card requests TC and CDA performance fails, the terminal rejects the transaction and returns a response code.
- The online confirmation result of the Issuer may become transaction rejection due to issuer authentication result and some options of the card.
- In the process of transaction processing, indicator and counter will reflect the occurring situation.
- After online authorization, indicator and counter may be reset according to issuer authentication result and card options.

The terminal can perform some additional functions to complete the whole transaction, such as printing transaction receipt and recording transaction data, etc., functions which do not conflict with EMV Integrated Circuit Card Specifications for Payment Systems, Version 4.3 Book 4

4.2.11.2 Card Data

For completing the processing, some of the data used in the card are as shown in Table 37. Please refer to *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part2: Debit/Credit Application Card Specification* for other data elements:

Table 37: Transaction Completion - Data Elements Used by Card

Data Element	Description
Application default action (ADA)	The indicator defined by the Issuer for designating card actions under some special conditions.
CDOL2	Listing the data objects (tag and length) the card requires the terminal to transmit in the second generate application cryptogram command. The following data in CDOL 2 are used in card risk management check: • Transaction currency code • Country code of the terminal • Authorized amount • Authorization response code Terminal verification result (TVR)

The data that the card responds to Generate AC command are as shown in the table below:

Table 35: Ending of Transaction - Generate AC Command Card Response Data

Data Element	Description
Application cryptogram (AC)	Cryptogram generated by the card
Application transaction counter (ATC)	Counter for the card to record number of transaction
Cryptogram information data	Including the following indicator bits: CRYPTOGRAM TYPE • Refusing AAC • Accepting TC • Submitting online ARQC Other status information
Issuer application data	Application data defined by Issuer, including CVR
Card verification result (CVR)	Indicating the offline processing result of current and last transaction

4.2.11.3 Terminal Data

The data elements used by the terminal in completing processing are as shown in Table 39:

Table 39 End Transaction - Data Used by Terminal

Data Element	Description
Authorization response code	Indicating transaction processing result and submitting it to card
Terminal verification result (TVR)	It is for recording offline processing result, such as SDA performance situation, etc.

4.2.11.4 Command

GENERATE AC

The terminal sends the second Generate AC command to the card for requesting the final application cryptogram, and the Generate AC command may also indicate that combined dynamic data authentication/ application cryptogram generation (CDA) is to be executed.

GENERATE AC command includes the terminal data elements described in details in CDOL 2 of card. The terminal obtains those data elements via reading application data. CDOL 2 data include the authorization response code returned online by the Issuer or the authorization response code returned by the terminal in case that it is unable to complete authorization online.

Command parameter P1 indicates the type of encryption requested by terminal, which is described in detail in Table B.7, Appendix B of UNIONPAY INTEGRATED CIRCUIT CARD SPECIFICATIONS - BASIC SPECIFICATIONS PART2: DEBIT/CREDIT APPLICATION CARD SPECIFICATION.

The response information of command GENERATE AC includes card transaction counter, cryptogram type indicating card authorization decision, application cryptogram and processing result designated by CVR. The self-defined data of the Issuer can also be returned.

4.2.11.5 Processing Flow

The terminal may process different situations during the period of completing processing according to the situation occurring in the previous transaction processing:

After card action analysis is ended, the card may have:

- Requested for offline approval (TC) or transaction rejection (AAC);
- Requested for online authorization (ARQC).

If online processing is conducted, online authorization may:

- have been successfully completed
- have not been completed due to the problem caused by the terminal or communication.

If the card responds to the first Generate AC command in card action analysis with a TC or AAC, the transaction is accepted or rejected offline.

The terminal must determine the final transaction result according to the result of combined dynamic data authentication (CDA) displayed in CID and TVR returned by the first Generate AC command response.

Table 40 End Transaction - Terminal Processing Result (Offline)

First Generate AC Return Result	CDA Processing result	Final Transaction Result
TC	CDA not executed or succeed.	Offline Approval
TC	CDA fails.	Rejected
ARQC	CDA fails	Request AAC in the second Generate AC command.
AAC	--	Rejected

If the first Generate AC command returns ARQC (request for online) during card action analysis:

1. The terminal issues the second Generate AC command to the card for requesting GENERATE AAC or TC because the terminal does not support or the online authorization is not completed due to other problem.
2. If online authorization is completed, the terminal issues the second Generate AC command to card for requesting TC (approval) or AAC (rejection) according to online authorization result. The terminal processes the transaction according to the following situations:

Table 41 Transaction Completion - Terminal Processing Result (Online)

Online Authorization result	Terminal Requests Data from	Card Return	Final Transaction Result
-----------------------------	-----------------------------	-------------	--------------------------

		the Card		
Incomplete		AAC	AAC	Rejection
		TC	TC/AAC	Approval/rejection
Completed	Pass	TC	TC or AAC	In the following two situations, the card returns AAC (rejection) and in other situations, the card returns TC (approval): 1. Issuer authentication fails while that the transaction is rejected in this situation is identified in ADA. 2. Issuer authentication is a mandatory item but is not executed; meanwhile, it is identified in ADA that the transaction is rejected in such a situation.
	Decline	AAC	AAC	Decline

4.2.11.6 Flow Chart

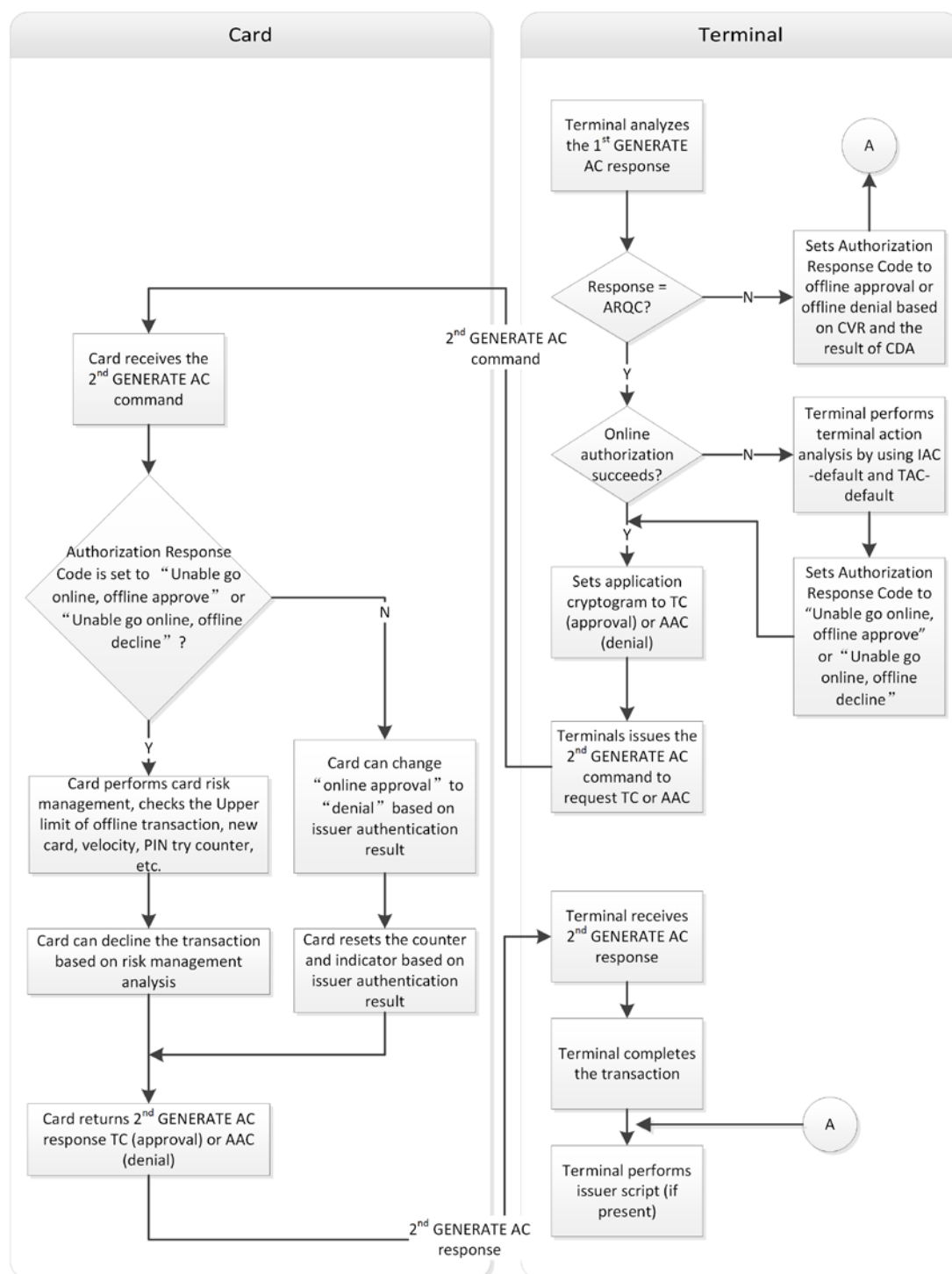


Figure 16 Flow Chart of Transaction Completion

4.2.11.7 Related Pre-Processing

Online Processing

If the card receives EXTERNAL AUTHENTICATE command sent by the terminal, the card begins to conduct issuer authentication processing while setting indicator as that

issuer authentication has already been performed and identifying its success or failure. The indicators will be used by the card in card response during the period of completing processing and to determine which card counters and indicators will be reset.

4.2.11.8 Related Post-Processing

Issuer Script Processing

If issuer command scripts are included in online processing scope message, the terminal will need to transmit the command scripts to the card to perform.

4.2.12 Issuer Script Processing

4.2.12.1 Description

Issuer script processing allows the Issuer to change personalization data of the card without re-issuing a card. The message returned by the Issuer in the authentication response includes the script of card commands. The terminal sends those commands to the card regardless of whether the security conditions are satisfied. For more information about issuer script processing details, please refer to the *UnionPay Integrated Circuit Card Specifications - Basic Specifications Part II: Debit/Credit Application Card Specification Appendix K*.

The supported script commands are as follows:

- Changing card parameters
- Blocking/unblocking application
- Blocking the card
- Resetting PIN counter
- Changing offline PIN

Issuer script processing prevents credit and fraud risks by blocking the stolen or maliciously overdrawn cards. In addition, card parameters can also change according to specific situation of the cardholder.

4.2.12.2 Script-related Key

4.2.12.2.1 Message Authentication Code Key (MAC KEY)

MAC key is used to generate and verify script MAC, which is the cryptogram included in the script command for confirming that the data have not been tampered with (integrity), and validate if the Issuer in the command is legal (issuer authentication). Three keys are used in MAC processing:

- MAC Master DEA key (MAC MDK):

It is the unique double length symmetric key determined by the Issuer for generating the unique message authentication code key (MAC UDK) of the card and the session key of transaction MAC.

- Card MAC UDK:

It is the double length symmetric key written into the card after derivation by the MAC Master DEA key during the card personalization. MAC UDK is used to generate MAC session key in transaction process.

- MAC session key:

MAC session key is the unique double length symmetric key in a transaction, and is used for generating the MAC code of script command.

4.2.12.2.2 Data Encryption Key

Data encryption key is used to encrypt the sensitive data in script such as offline PIN, etc. Three keys are used in data encryption:

- Master Data Encryption DEA Key (ENC MDK):

Data encryption master DEA Key is the unique double length symmetric DEA key of the Issuer, and is used to generate the unique data encipherment DEA card key and the data encryption session transaction key.

- Unique Data Encryption DEA Key (ENC UDK):

ENC UDK is the double length symmetric DEA key written into the card after the derivation of master data encryption DEA Key during card personalization.

- Data Encryption Session Key:

Data encryption session key is the unique double length symmetric key in transaction, and is obtained from ENC MDK derivation and used for the issuer host system to encrypt the sensitive data in the script.

4.2.12.3 Card Data

The counters and indicators used in the card in script processing are as shown in Table 34:

Table 34: Issuer Script Processing - Counters and Indicators Used by Card

Data Element	Description
Application transaction counter (ATC)	Transaction counters processed after the card is personalized are used in terminal frequency checking.
Card verification result (CVR)	Verification result indicator is set according to the offline processing result of the current and last transaction.

Issuer script command counter	Quantity of commands in security message received by the card after recording the second generated application cryptogram. It may be reset in the transaction completion processing for the next transaction.
Issuer script failure indicator	If script command fails, it indicates position “1” and the possible failures are: • Error in security message; • Security message passes but the command execution fails. • Security message is required but there is no such security message. It may be reset in the transaction completion processing for the next transaction.

4.2.12.4 Terminal Data

The data elements used by the terminal in issuer script processing are as shown in Table 35:

Table 35: Issuer Script Processing - Data Element Used by Terminal

Data Element	Description
Issuer script result	Recording the result of issuer script command processing and the result should be included in clearing message and the next online authorization.
Terminal verification result (TVR)	Two relevant indicator are included in TVR: • Script processing fails before the final GENERATE AC command. • Script processing fails after the final GENERATE AC command. <i>UnionPay Integrated Circuit Card Specifications - Basic Specifications</i> only supports processing issuer script after the last application cryptogram is generated.
Transaction status information (TSI)	TSI includes a tag indicating performance of issuer script processing.

4.2.12.5 Online Response Data

Table 36 Issuer Script Processing - Online Response Data

Data element	Description
Issuer script command	Every issuer script command in script is based on BER-TLV format, beginning with tag '86'.
Issuer script identifier	The unique identifier used by the Issuer to identify issuer script.
Issuer script template 2	<i>UnionPay Integrated Circuit Card Specifications - Basic Specifications</i> only supports issuer script template 2. Tag '72' identifies template 2; and the special script data of Issuer transmitted to the card after application cryptogram command is generated for the second time is included in the template.

4.2.12.6 Command

4.2.12.6.1 Application Block

The command will block the currently selected application. If an application is blocked in the transaction process, the card and the terminal will continue to process the transaction through processing completion. After the application is blocked, the card will reject the blocked application in completing any financial transaction. The terminal can select the locked application for unblocking the application.

4.2.12.6.2 Application Unblock

This command unblocks the blocked application. For an Issuer, it is best to conduct application unblocking through special designated devices.

4.2.12.6.3 Card Block

Card block will make all applications on card permanently locked.

4.2.12.6.4 PIN CHANGE/UNBLOCK

PIN CHANGE/UNBLOCK command allows the Issuer to change the PIN of the card while unblocking the PIN (resetting PIN retry counter). Change/unblock PIN should be conducted in an environment according to the Issuer's security requirement.

4.2.12.6.5 Put Data

The parameters of PUT DATA command should be used to update the management parameters set by the Issuer in a card, such as upper limit of consecutive offline transaction times, lower limit of consecutive offline transaction times, limit on consecutive offline

international transactions and the upper limit of total amount of accumulated offline transactions, etc.

4.2.12.6.6 Update Record (UPDATE RECORD)

The UPDATE RECORD command modifies the content of a record.

4.2.12.7 Processing Flow

Issuer script processing includes three aspects: issuer script, command execution, and security message.

4.2.12.7.1 Issuer Script

The Issuer sends issuer script to the Acquirer via a return message.

If the returned message includes tag '72', it means that issuer script should be performed after the final Generate AC.

4.2.12.7.2 Command Execution

The issuer script command is used in modifying the personalized data in the card. The card will execute the requested command to update the data included in the card only if the script command supports security message and the security message is successfully performed. Before processing the issuer script command, the Issuer should first successfully perform issuer authentication using security message. The Issuer undertakes issuer script command organization. If an entity different from the Issuer initiates a script command, it also must execute security authentication.

4.2.12.7.3 Security Message

The principle purpose of security message is to ensure the confidentiality of data, integrity of messages and issuer authentication. The confidentiality of data ensures that the secret data will be kept confidential during transmission from the Issuer to card. The integrity of messages ensures that the command and command data will not be changed during transmission. Issuer authentication ensures that the command is from a valid Issuer. MAC is used to achieve integrity of messages and issuer authentication. The confidentiality of data is achieved via encrypting plaintext command data (if the data appears).

4.2.12.8 Flow Chart

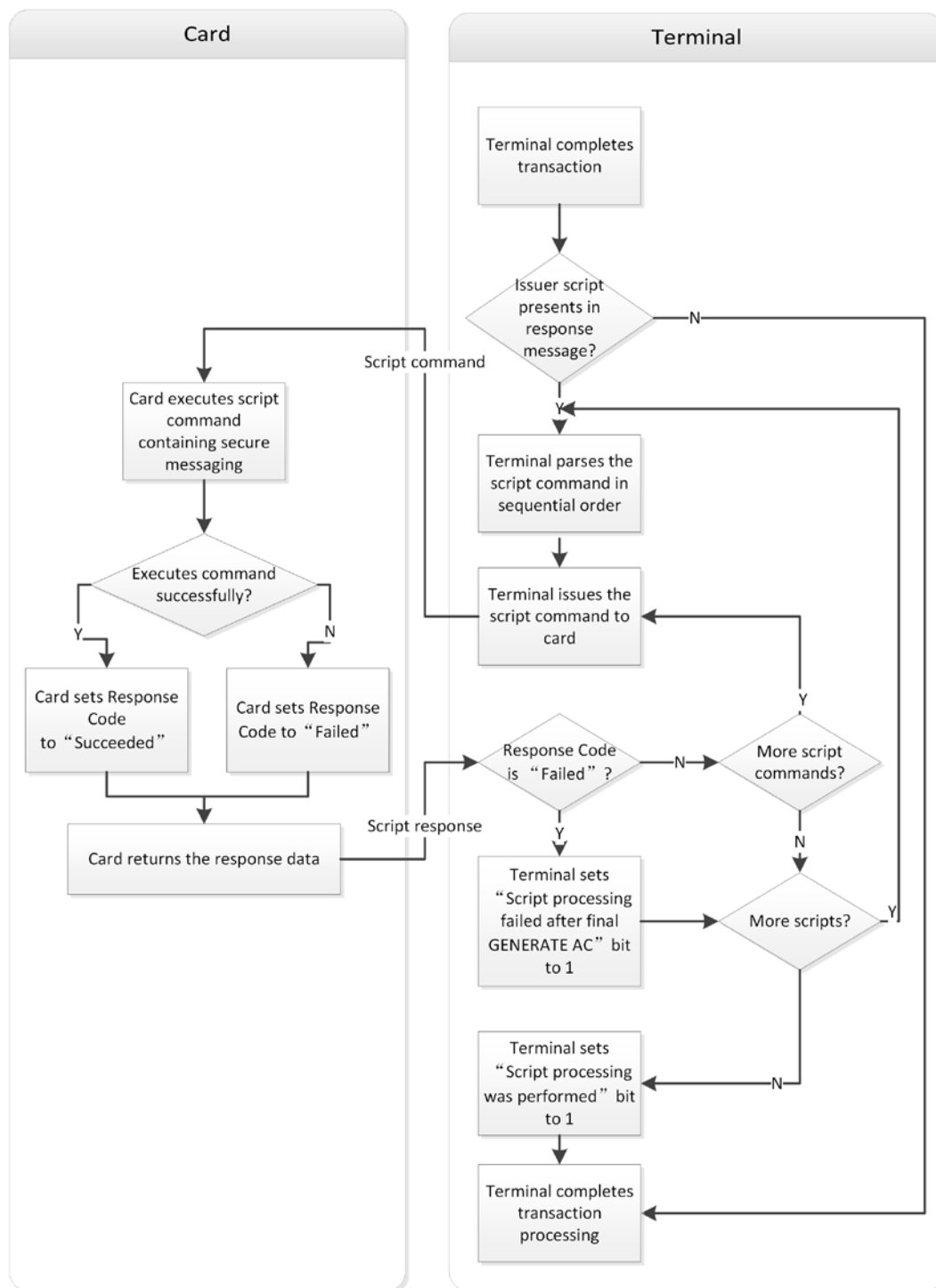


Figure 17 Flow Chart of Issuer Script Processing

4.2.12.9 Related Pre-Processing

Online Processing

Online processing response message may include issuer script that is required to be processed in issuer script processing.

4.2.13 Card Transaction Detailed Log

4.2.13.1 Description

A card may support the transaction detailed log. For a card that supports transaction log, the response to SELECT command should contain Log Entry data element while also supporting the terminal to obtain Log Format data element from the card via GET DATA command. The terminals requiring access to transaction detailed log records can send READ RECORD command to the card to read transaction records one by one.

A card that supports record of transaction detailed log should obtain terminal data elements required for recording transaction details from the terminal via DOL. When the transaction ends, if the card approves the transaction and returns TC, the transaction details will be recorded in the card for the cardholder to query offline.

Transaction detailed log is stored in a file on the card in the form of circular record file and the modification of the file of the transaction detail should be completed within the card. The terminal can only conduct read operations.

4.2.13.2 Data Elements for Transaction Details (Recommended)

It is suggested that the transaction detailed log of the card include the data such as transaction date, transaction time, authorized amount, other amounts, terminal country code, transaction currency code, merchant name, transaction type and application transaction counter, etc. For the detailed format, please see Table 42:

Table 42 Card Transaction Details - Data Format

Data	Format	Length (byte)
Transaction date	YYYYMMDD	3
Transaction time	HHMMSS	3
Authorized amount	N12	6
Other amounts	N12	6
Terminal country code	N3	2
Transaction currency code	N3	2
Merchant name	ans	20

Transaction type	N2	1
Application transaction counter(ATC)	b	2

5 Security, Key, and Digital Certificate

This section defines the contents related to security in the process of debit/credit application and includes the following:

- Offline static data authentication
- Offline dynamic data authentication
- AC generation and issuer authentication
- Security message

Please refer to UnionPay Integrated Circuit Card Specifications - Basic Specifications Part4: Debit/Credit Application Security Specification for the detailed requirements concerning security in the various steps and implementation guidelines.

6 AID Reservation and Usage

6.1 Application Identifier (AID) Structure and Length

This Specification uses the Application Identifier (AID) basic structure specified in ISO/IEC 7816-5, consisting of application identifiers (AID) that is 5-16 bytes in length, plus a 5 byte registered application provider identifier (RID) and a 0-11 byte proprietary application identifier extension (PIX).

Application identifier (AID) is the unique identifier of card applications. Requesting organization submits request to financial industry authority, which then organizes and allocates in centralized fashion.

For application type code definitions please see Table 42.

6.2 Application Provider Identifier (AID) Overall Encoding

Application Provider Identifier (RID) overall encoding follows ISO/IEC 7816-5 rules regarding AID, application identifier (AID) encoding is as follows:

Table 43 Application Identifier (AID) Encoding

Application Provider Identifier (RID)	Proprietary Application Identifier Extension (PIX)
5 bytes	0 - 11 bytes

6.3 Proprietary Application Identifier Extension (PIX) Encoding

For specifics please see Table 44.

Application Type Code must be present, for definitions please see Table 42.

Reserved field is optional, if it is not there, the field that follows should also not be present. If that field is present, then the value of the field would be up to the Issuer to define.

Proprietary definition identifier is optional. If it is not present, the proprietary definition field should not be present. If this field is present, see Table 45 for its definition.

Proprietary definition field is optional.

Table 44 Proprietary Application Identifier Extension (PIX) Encoding

Application Type Code	Reserved Field	Proprietary Definition Identifier	Proprietary Definition Byte

3 bytes	0-1 bytes Reserved for Issuer	0-1 bytes 00 Indicates that the Proprietary Definition Byte content has the reserved definition in this Specification 01 Indicates that the Proprietary Definition Byte content is defined by the mobile payment application 02-FF Reserved	0-6 bytes
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6.4 Application Type Code Definition

The first byte of Application Type Code with value 00-FF is application type codes reserved for use in this book. For partial definition of values see Table 45. The value FF for the first byte of the Application Type Code is reserved for the Issuer.

Table 45 Application Type Code Definitions

Value	Definition
01 01 01	Debit
01 01 02	Credit
01 01 03	Quasi Credit